

Chiral Symmetry on the Lattice

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1 Introduction

Lattice quantum mechanics is one of, if not the most, successful schemes for the study of physical matter in history, both in terms of theoretical understanding and of phenomenology. This fact holds equally well for tangible quantum matter (semiconductors, magnetic materials, photonic devices, etc) as it does for exotic quantum matter (quark-gluon condensate, neutron star cores, superconductors, etc). Lattice QM is a cornerstone of condensed-matter physics and QCD.

The essence of lattice QM is the use of finite-dimensional Hilbert spaces for modelling matter through the discretisation of space.¹ This can either be a reflection of some real crystalline nature of a solid under study, as is the case for solid-state physics, or it can be used as an approximation scheme for a continuous medium. This discretisation renders difficult theories tractable, either via the existence of lattice-only analytic solutions, or as facilitated by modern computing methods. Lattice QM is simulated quite easily on computers; it would be fair to say that supercomputers employing lattice QCD are responsible for the most precise numerical understanding of QCD to date.

The lattice description naturally differs from continuum theories in many ways, most obviously in the inability to describe spatial derivatives on a lattice. This, perhaps surprisingly, is not a real issue when it comes to describing continuum theories using lattice QM. This is not to say the scheme doesn't come with its own obstacles; most notable for our work is the very subtle obstruction provided by the *Nielsen-Ninomiya theorem*. The upshot of this theorem is the statement:

Theorem 1.1. *Nielsen-Ninomiya Theorem (Simplified):* *For every right-moving particle in a Hamiltonian lattice theory, there is a left-moving copy.*

This idea will play a central rôle in our story moving forward. Nielsen-Ninomiya all but forbids putting chiral gauge theories on the lattice; for which the left- and right-handed² particles are charged differently under the gauge symmetries of the theory, the most prominent example being the standard model. Nielsen-Ninomiya also limits our ability to emulate *chiral anomalies* on the lattice.³ The chiral anomaly is associated with the non-vanishing divergence of an axial current $\partial^\mu j_\mu^A \neq 0$. Interpreting the axial charge Q^A as $(\# \text{left-moving fermions}) - (\# \text{right-moving fermions})$, the chiral anomaly becomes a statement that the relative abundance of left/right-moving fermions changes, [Kad13] something which is forbidden under Nielsen-Ninomiya.

The novelty of the work by Chatterjee, Pace, and Shao [CPS25] early last year is their construction of a particular $U(1)$ charge Q^A in a lattice theory for the *Schwinger model*. The Schwinger model [Sch62] is a theory of $1 + 1$ -dimensional massless QED, in a sense the simplest fermionic gauge theory. This lattice charge operator Q^A flows to the continuum charge operator Q^A in the continuum limit of the theory, and in this way becomes our lattice analogue for the axial charge. The story of how this charge operator behaves can be summarised in three points:

- (a) In the continuum theory, we have the relation $Q^A = \mathcal{C}^R Q^V (\mathcal{C}^R)^{-1}$, where $\mathcal{C}^R$ represents the right-moving charge conjugation operator, and Q^V is the vector charge operator in the continuum. We know how to define Q^V , the lattice analogue of Q^V , so we find a lattice analogue for $\mathcal{C}^R$ and define Q^A accordingly.

¹And, in the case of lattice QCD, time.

²We use the terms “left/right-handed” and “left/right-moving” interchangeably, but we recognise that the latter is typically only used to refer to theories of $1 + 1$ -dimensions. Furthermore, our entire discussion is limited to theories of odd spatial dimensions, as chirality is not a sensible notion otherwise.

³As is noted in [CPS25], the primary barrier to recognising perturbative anomalies, like the chiral anomaly, on the lattice is really the finite dimension of our Hilbert space; our trace-class operators are simply matrices.

- (b) The axial charge Q^A does not commute with the vector charge Q^V , $[Q^V, Q^A] \neq 0$. In this way Nielsen-Ninomiya is not violated, but we seemingly compromise our lattice analogue because in the continuum, Q^V and Q^A are independent.
- (c) On the contrary, this non-commutativity generates the *Onsager algebra* [Ons44], the non-locality of which will mean we cannot gauge the mixed symmetry of Q^V and Q^A . This obstruction to gauging flows to match the chiral anomaly of the continuum theory.

In what follows, we expound the Chatterjee-Pace-Shao paper. We begin with a whirlwind explanation of Hamiltonian lattice QM, and go on to explain and justify the key steps of the paper.

2 Prerequisites for Hamiltonian lattice theories

Lattice quantum mechanics is one of the foundational tools of the condensed matter theorist, and contains within its formalism all of the most famous toy models like the Ising, XX, and Hubbard models. Even the simplest theories of solid matter off-lattice, like jellium, have immediate complications which require usage of brutal approximation schemes.⁴ The theory is a d -dimensional discrete spatial lattice formulation, in which time is an additional continuous variable which parametrises unitary evolution; any attempt to make Poincaré symmetry exact in this formulation is something of a non-starter. As their many-body nature insists on the usage of multi-particle states, these theories make heavy use of second-quantisation. As such, when we write down a Hamiltonian $\hat{H}$ to dictate our theory we are forced to write it in terms of second-quantised field operators.

Remark 2.1. Looking at Hamiltonian lattice QM from the gallery, it mightn't be at all obvious what the analogue of, say, a Lagrangian kinetic energy term $\bar{\psi}\not{\partial}\psi$ would be in this formalism, in particular given the fact that we cannot define spatial derivatives for fields on a discrete space.⁵ We give a brief guide to common terms in these Hamiltonians in [appendix A](#). One property we define explicitly here is that of *ultra-locality*: an operator is said to be ultra-local precisely when it couples the fields at points which differ by some fixed distance which does *not* scale with the system size.

In analogy with continuum QFT, the vacuum state $|\text{vac}\rangle$ of a free lattice theory should be interpreted as a lattice devoid of particles, but with one bonus: whereas the infinitude of the continuum theory does not allow for a robust Fermi-sea interpretation of vacua, the finiteness of lattice allows for exactly that. We will put this to use later where the vacuum $|\Omega\rangle$ of an interacting theory may be viewed as a certain configuration of states filling the negative energy modes of the lattice.

We are interested in these lattice theories for their application as a regularisation scheme of quantum field theories. Many continuum QFTs emerge from lattice theories as the low-energy effective field theories in the infinite-extent lattice, i.e., where we take the volume of our theory to infinity while restricting our attention to the low-energy modes. Apart from the obvious computational tractability gained by dealing with a finite system, these lattice regularisations impose UV and IR cut-offs. For a homogenous, isotropic lattice of $N_1 \times N_2 \times \dots \times N_d$ sites with lattice spacing a , we have a baked-in UV cut-off given by $\Lambda_{\text{lattice}} \sim \frac{1}{a}$, as below a wavelength of $\sim a$, aliasing becomes incredibly problematic⁶. Additionally, in order to satisfy reasonable boundary conditions⁷, wavelengths cannot be on or greater than the order of $\sim a \min(\{N_1, \dots, N_d\})$, whence a baked-in IR cut-off emerges.

⁴See: Bogolyubov's hunting license [Bog47] for an example.

⁵At least we cannot define spatial derivatives *as such*.

⁶We take the term aliasing here from the world of sound processing. An example is given very shortly in subsection 2.2.

⁷Periodic, anti-periodic, etc.

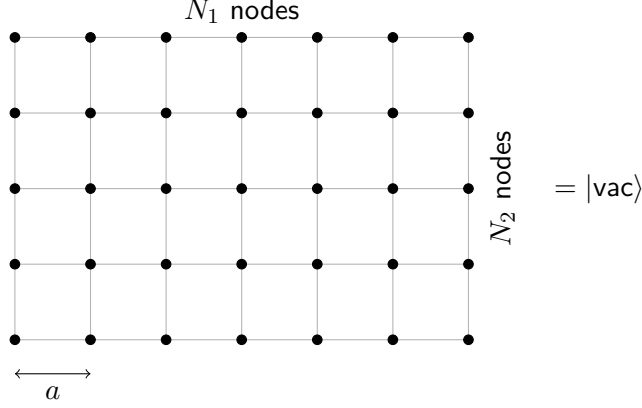


Figure 1: Empty homogeneous isotropic two-dimensional lattice with spacing a .

Remark 2.2. It should be noted that there is something of an art to finding these lattice regularisations: rather than being able to produce automatically the “lattice version” of a continuum theory through discretisation, our actual process is that of writing down lattice theories and seeing what they look like in the IR, infinite-extent limit. This is analogous to the monotonicity of RG flow; we can only flow towards an IR theory, we cannot flow towards the UV.

In this work, we will be working with fermionic theory of a 1-dimensional lattice of L nodes.⁸ This low-dimensionality will simplify the discussion immensely, and will enable us to study the lattice analogue of the *Schwinger model*, [Sch62] the theory of massless QED in $1 + 1$ -dimensions. We expound on all of the above in the remainder of this section.

2.1 Lattice units

To find the continuum theory in the IR of the infinite-extent lattice, it is necessary to properly establish a separation of scales through the introduction of a tunable UV cut-off. We must therefore define our units and understand the implications they have on our labelling of energy and momentum. Following the conventions of [CPS25], we opt to fix $a = 1$ and use the $L \rightarrow \infty$ limit, which has the advantage of being able to label our lattice sites by integer indices $1, \dots, L$.

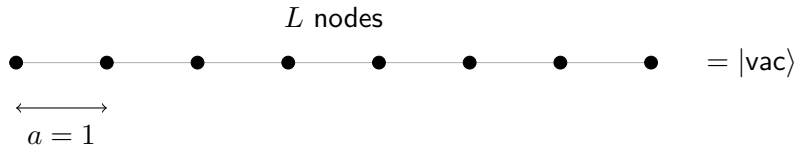


Figure 2: One-dimensional lattice with L sites and unit spacing.

We can use the exact same indices to label the reciprocal lattice, where we define our Fourier transforms⁹ like so:

$$\tilde{f}(k) := \sum_{j=1}^L e^{2\pi i j k / L} f(j).$$

⁸For convenience, we assume even $L \in 2\mathbb{N}$, but insofar as we understand it, this assumption is harmless.

⁹In this context, these are DFTs: *discrete Fourier transforms*.

This convention differs from natural units: space and momentum now have the *same* units¹⁰, rather than behaving reciprocally; our Brillouin zone is no longer written as $[0, 2\pi/a)$, but rather as the interval $(0, L]$.¹¹ In the above units, our baked-in lattice UV cut-off $1/a$ is simply 1, and hence our separation of scales looks like:

$$\frac{1}{L} \ll E \leq \Gamma \ll 1,$$

where we've introduced the tunable UV cut-off Γ ; which will allow us to restrict to the infrared spectrum to study the emergent continuum theory.

2.2 Why is the continuum in the IR anyway?

A reader may reasonably wonder why we should expect to find a continuum theory as a low-energy EFT of a lattice theory in the first place. We invite such a thoughtful reader to imagine a line of evenly-spaced buoys bobbing up and down, dotted across the length of a swimming pool. If somebody jumps into the pool causing a great splash, the buoys will bob up and down sporadically, completely out of sync with their neighbouring buoys. This is the rearing of aliasing; we could be forgiven for looking at the configuration of the buoys at some fixed time and concluding that they are representative of some low-energy wave.

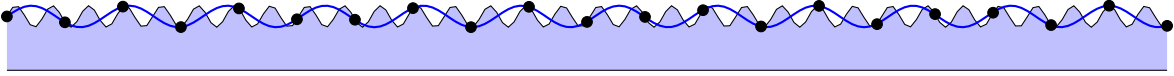


Figure 3: A choppy wave. The blue sinusoid is the *alias* low-energy wave.

Whereas, if a very gentle wave spreads across the water, with wavelength comparable to the pool's length, the buoys will rise and fall smoothly alongside their neighbours. These buoys are our spatial indices and can be labelled from $1, \dots, L$.

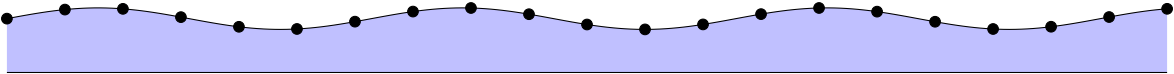


Figure 4: A gentle wave.

If we extend the buoys out farther, and make lower-energy excitations in the water, the collective motion of the buoys captures the continuous motion of the wave better and better. The reader should keep this analogy in mind when considering everything that follows.

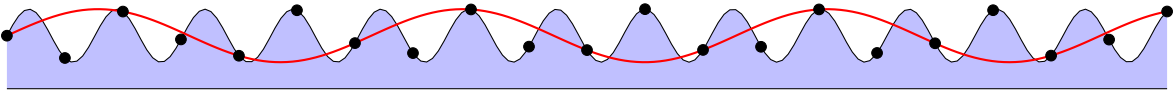


Figure 5: A left-mover on the lattice. The red sinusoid represents the physical momentum of the wave. Multiplying it by the wavelength 2 oscillation yields the water wave.

Later on we will find that, while the low-energy waves will correspond directly to right-moving fermions in the continuum, the left-moving fermions will correspond to a rapid oscillation with wavelength 2 multiplied by a low-energy wave; effectively flipping the sign of the field on each neighbouring lattice site, setting each buoy in counter-step to its neighbours. It is at this point that we must admit our buoy analogy breaks down.

¹⁰We use the word *unit* liberally here. If you prefer, it is only our labelling system that changes.

¹¹We use the π/a -centred Brillouin zone here rather than the 0-centred $[-\pi/a, \pi/a)$ interval.

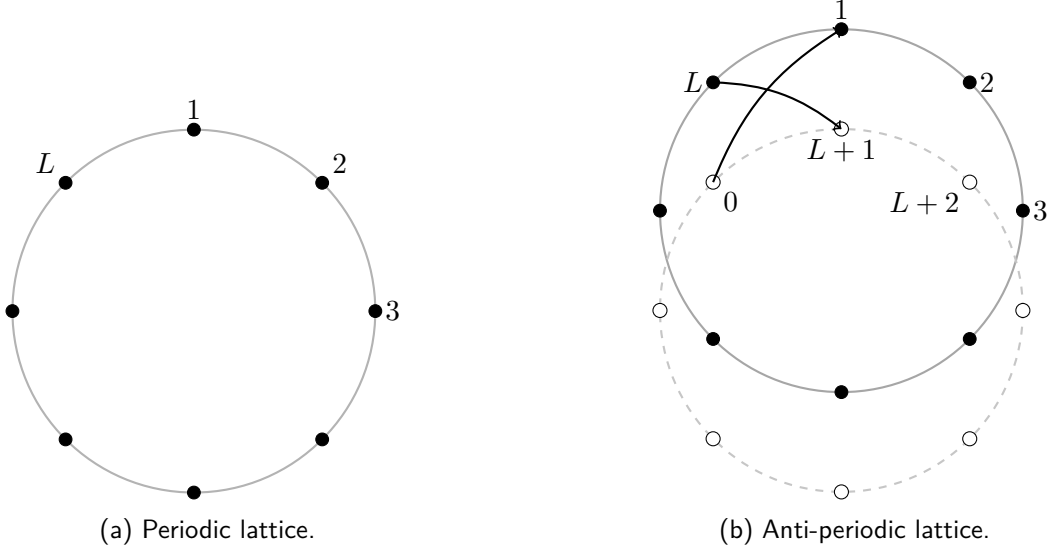


Figure 6: Lattice boundary conditions: In the anti-periodic case, the dotted loop indicates the periods $(nL, nL + L]$ for n odd. We witness the anti-periodic lattice as a double cover of the periodic lattice.

2.3 Second-quantised formalism

Recall that we imagine the vacuum of a free lattice theory to be a lattice devoid of particles. We take this state to have normalisation $\| |\text{vac}\rangle \|^2 = |\langle \text{vac} | \text{vac} \rangle| = 1$. For our purposes we have one fermion field c_j , where j indicates the site index at which a fermion is created when the operator $c_j^\dagger$ acts on the vacuum as $c_j^\dagger |\text{vac}\rangle = |j\rangle$, with normalisation $\| |j\rangle \|^2 = |\langle j | j \rangle| = 1$. We allow j to range over all of $\mathbb{Z}$, but we enforce periodic/anti-periodic boundary conditions given by the constraint $c_{L+j} = (-1)^\nu c_j$ for $\nu \in \{0, 1\}$. As such we only have L independent indices.

In order to respect our periodicity, we introduce the periodic Kronecker delta function $\delta_j := \sum_{k \in \mathbb{Z}} \delta_{j, Lk}$, differentiated from $\delta_{i,j}$ by its bold font and single-indexing.¹² A simple way to think about this operator is with its piecewise definition:

$$\delta_j = \begin{cases} 1 & j = 0 \pmod L \\ 0 & \text{else.} \end{cases}$$

We will also need the (possibly) anti-periodic Kronecker delta $\delta_j^\nu := \sum_{k \in \mathbb{Z}} (-1)^{\nu k} \delta_{j, kL}$, which has its own piecewise definition as well:

$$\delta_j^\nu = \begin{cases} 1 & j = 0 \pmod{2L} \\ (-1)^\nu & j = L \pmod{2L} \\ 0 & \text{else.} \end{cases}$$

Of course, for $\nu = 0$, these two distributions are equal. Our fermion operators satisfy the canonical anti-commutation relations $\{c_j, c_{j'}\} = 0$, $\{c_j^\dagger, c_{j'}^\dagger\} = 0$, $\{c_j^\dagger, c_{j'}\} = \delta_{j-j'}^\nu$.

¹²This is the finite version of the Dirac comb $\text{III}(x)$.

2.4 The Nielsen-Ninomiya theorem and fermion doubling

In order to properly understand the restriction of Nielsen-Ninomiya on chiral fermions on the lattice, we need to formulate the theorem more precisely.

Theorem 2.1. The Nielsen-Ninomiya theorem: Let $\hat{H}$ be a lattice Hamiltonian which:

- (a) Is built of fermion bilinears;
- (b) Commutes with the vector charge Q^V ;
- (c) Commutes with the lattice translation operator $\hat{T}_{\hat{\mu}i}$ ¹³;
- (d) Is local, meaning continuous in crystal momentum.

Then, within each irreducible representation of the global symmetries of $\hat{H}$, there must be an equal number of left- and right-moving fermions, where we fixed a specific charge sector of the global symmetries.[CPS25],[Kad13],[Fri82].

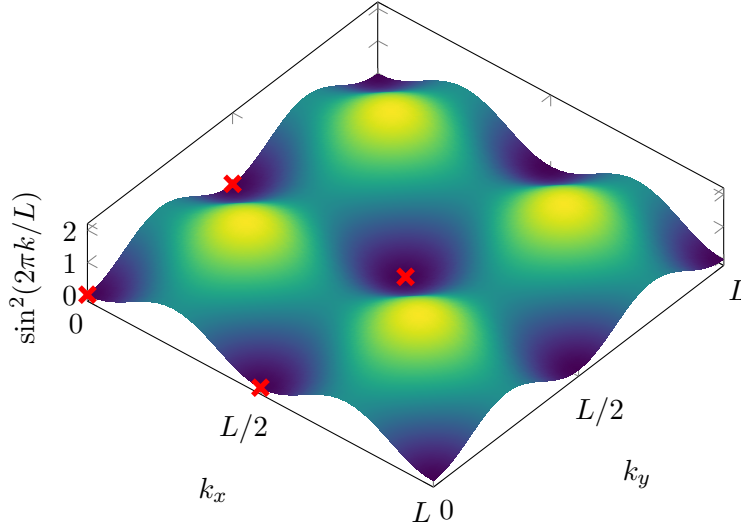


Figure 7: Surface plot of $\sin^2(2\pi k/L)$ on $[0, L] \times [0, L]$.

How should this theorem be understood practically? As we will see in section 4, the story is told in the spectrum of the theory. For now, we draw a comparison between the correlation functions found in a continuum theory versus those found in a lattice theory, where we will witness *fermion doubling*: a lattice artifact whereby each fermion has a doppelgänger for each spatial dimension of the theory. So in $1 + 1$ -dimensions, we find pairs of left-moving and right-moving fermions, and in $1 + 3$ -dimensions, we find fermion quadruplets (the x -direction doppelgänger also has its own y -direction doppelgänger!). The following comparison is from a Euclidean-signature Lagrangian lattice theory where time has also been discretised, and as such won't apply directly to our Hamiltonian theories, but should help elucidate fermion doubling for now. Recall the $1 + 3$ -dimensional massless fermion propagator, familiar from continuum QFT:[Ton06]

$$S_F(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i \not{p}}{p^2 + i\epsilon} e^{-ip(x-y)}.$$

¹³Definition in [appendix A](#).

Compare this to the lattice analogue:[Rot92]

$$S_{\text{Lattice}}(x-y) = \lim_{L \rightarrow \infty} \int_{\text{BZ}} \frac{d^4 k}{(2\pi)^4} \frac{i\gamma^\mu (\sin 2\pi k_\mu / L)}{\sin(2\pi k_\nu / L) \sin(2\pi k_\nu / L)} e^{-ik(x-y)},$$

where the integral is performed over the Brillouin Zone BZ.¹⁴ In the infinite-extent limit, we may hope that only the values around $k = 0$ dominate this integral meaning we can effectively replace $\sin k$ ¹⁵ with k , and in fact this works perfectly for scalar fields. However, for our fermions, we must grapple with the fact that there are *sixteen* distinct points in the BZ for which $\sin 2\pi k/L$ is small, i.e. all those points for which $k_\mu \sim 0$ or $k_\mu \sim L/2$. See figure 7 for an example of these points for a theory in 2-dimensions. Note that the unmarked valleys are actually periodic copies of the valleys that have already been marked.

3 1 + 1D free massless Dirac fermions in the continuum and on the lattice

Having painted a birds-eye view of lattice QM, it falls on us to get into the details of the Chatterjee-Pace-Shao paper. Ultimately we will be discussing the Schwinger model, but we will benefit immensely from understanding the ungauged theory of free massless Dirac fermions in 1 + 1-dimensions as a basis upon which the lattice gauge theory can be built. We begin with an explanation of the symmetries of the continuum theory from which to draw the lattice analogy.

3.1 In the continuum

Our continuum theory is given by:

$$S = i \int dt dx \bar{\Psi} \Gamma^\mu \partial_\mu \Psi,$$

where $\eta_{\mu\nu} = \text{diag}(1, -1)$ and $\bar{\Psi} := \Psi^\dagger \Gamma^0$ where $\{\Gamma^0, \Gamma^1\}$ is our 1 + 1 Clifford algebra which we express via the Pauli matrices as: $\Gamma^0 := \sigma^x$, $\Gamma^1 := -i\sigma^y$, where Ψ is acted on as a column vector with Weyl-spinorial components $\begin{pmatrix} \Psi_R \\ \Psi_L \end{pmatrix}$. Recalling that $(\sigma^x)^2 = \mathbb{1}$ and $-i\sigma^x\sigma^y = \sigma^z$, we find:

$$\begin{aligned} \bar{\Psi} \Gamma^\mu \partial_\mu \Psi &= \Psi^\dagger \Gamma^0 \Gamma^0 \partial_0 \Psi + \Psi^\dagger \Gamma^0 \Gamma^1 \partial_1 \Psi = \Psi^\dagger \partial_t \Psi + \Psi^\dagger \sigma^z \partial_x \Psi \\ &= \Psi_R^\dagger \partial_t \Psi_R + \Psi_L^\dagger \partial_t \Psi_L + \Psi_R^\dagger \partial_x \Psi_R - \Psi_L^\dagger \partial_x \Psi_L \\ &= \Psi_L^\dagger (\partial_t - \partial_x) \Psi_L + \Psi_R^\dagger (\partial_t + \partial_x) \Psi_R. \end{aligned}$$

Hence the action is decoupled into a left and right part: $S = i \int dt dx \left[\Psi_L^\dagger (\partial_t - \partial_x) \Psi_L + \Psi_R^\dagger (\partial_t + \partial_x) \Psi_R \right] =: S_L + S_R$. Focusing on S_L for now, we can identify two global symmetries, namely charge conjugation and a global phase symmetry, as demonstrated below.

Proof. Invariance under a $U(1)^\mathcal{L}$ symmetry for $\phi \in \mathbb{R}/2\pi\mathbb{Z}$ where $\begin{cases} \Psi_L \mapsto e^{-i\phi} \Psi_L \\ \Psi_L^\dagger \mapsto e^{+i\phi} \Psi_L^\dagger \end{cases}$:

$$S_L \mapsto i \int dt dx e^{+i\phi} \Psi_L^\dagger (\partial_t - \partial_x) \Psi_L e^{-i\phi} = e^{+i\phi - i\phi} S_L = S_L.$$

□

¹⁴If you're wondering, correctly, how k_0 fits in the Brillouin zone, this equation comes from an isotropic hypercubic lattice where BZ is a stand-in for $(0, L] \times \dots \times (0, L]$.

¹⁵By $\sin k$ in this case we mean $\sqrt{\sin k_\mu \sin k^\mu}$.

We can write the $U(1)^\mathcal{L}$ symmetry in the operator form by defining a charge operator $\mathcal{Q}^L$ via $[\mathcal{Q}^L, \Psi_L] = -\Psi_L$, such that Ψ_L lives in the charge sector with charge -1 . Identifying this definition as the statement that Ψ_L is an eigenvector of $\text{ad}_{\mathcal{Q}^L}(\Psi_L)$ with eigenvalue -1 , we can employ Campbell's lemma to find:

$$e^{i\phi\mathcal{Q}^L}\Psi_L e^{-i\phi\mathcal{Q}^L} = \text{Ad}_{\exp i\phi\mathcal{Q}^L}(\Psi_L) = \exp(i\phi\text{ad}_{\mathcal{Q}^L})(\Psi_L) = e^{-i\phi}\Psi_L.$$

Likewise, $\Psi_L^\dagger$ has charge $+1$ under $\mathcal{Q}^L$.

Remark 3.1. Throughout this work, we will be making extensive use of $U(1)$ charge operators. In particular, we will be frequently looking at the integrality¹⁶ of the spectrum of operators. The reason we care about this is simply that if an operator Q has integer eigenvalues, so must ad_Q and therefore e^{iQ} is a valid representation of the group $U(1)$.

We define the charge conjugation operator via $\mathcal{C}^L \mathcal{Q}^L (\mathcal{C}^L)^{-1} = -\mathcal{Q}^L$, with the constraint that $\mathcal{C}^L \Psi_L (\mathcal{C}^L)^{-1} = \Psi_L^\dagger$.

Proof. Invariance under $\mathbb{Z}_2^{\mathcal{C}^L}$ generated by $\mathcal{C}^L$ where $\begin{cases} \Psi_L \mapsto \mathcal{C}^L \Psi_L (\mathcal{C}^L)^{-1} = \Psi_L^\dagger \\ \Psi_L^\dagger \mapsto \mathcal{C}^L \Psi_L^\dagger (\mathcal{C}^L)^{-1} = \Psi_L \end{cases}$:

$$\begin{aligned} S_L &\mapsto i \int dt \, dx \, \Psi_L (\partial_t - \partial_x) \Psi_L^\dagger = -i \int dt \, dx \, [(\partial_t - \partial_x) \Psi_L^\dagger] \Psi_L \\ &= -i \int dx \left(\int dt (\partial_t \Psi_L^\dagger) \Psi_L \right) + i \int dt \left(\int dx (\partial_x \Psi_L^\dagger) \Psi_L \right) \\ &= i \int dx \left(\int dt \Psi_L^\dagger \partial_t \Psi_L \right) - i \int dt \left(\int dx \Psi_L^\dagger \partial_x \Psi_L \right) \\ &= i \int dt \, dx \, \Psi_L^\dagger (\partial_t - \partial_x) \Psi_L = S_L, \end{aligned}$$

where above we have assumed that boundary terms vanish when applying integration by parts.¹⁷ $\square$

The symmetries $U(1)^\mathcal{L}$ and $\mathbb{Z}_2^{\mathcal{C}^L}$ together generate the larger global symmetry group $O(2)^\mathcal{L}$, for which the $U(1)^\mathcal{L}$ will be of particular interest to us. We find the same relations for $O(2)^\mathcal{R}$, $\mathcal{Q}^R$, and $U(1)^\mathcal{R}$. The proof of this can be found in [appendix D](#). Because these two $O(2)$ symmetries are independent, we have the vanishing commutator $[\mathcal{Q}^L, \mathcal{Q}^R] = 0$, and that our global symmetry of S can be written as a group direct product $O(2)^\mathcal{L} \times O(2)^\mathcal{R}$, which should be reproduced in the continuum limit of our lattice theory. We focus largely on the connected component of this global symmetry generated by the $U(1)^\mathcal{L}$ and $U(1)^\mathcal{R}$ symmetries. We define the vector and axial charges as

$$\mathcal{Q}^V := \mathcal{Q}^R + \mathcal{Q}^L \text{ and } \mathcal{Q}^A := \mathcal{Q}^R - \mathcal{Q}^L,$$

respectively. Notice that, because $\mathcal{C}^L \mathcal{Q}^L (\mathcal{C}^L)^{-1} = -\mathcal{Q}^L$ and $\mathcal{C}^L \mathcal{Q}^R (\mathcal{C}^L)^{-1} = \mathcal{Q}^R$, we could also define the axial charge like so:

$$\mathcal{Q}^A = \mathcal{C}^L \mathcal{Q}^V (\mathcal{C}^L)^{-1}. \quad (\spadesuit)$$

We keep this in mind for later when we discuss the EFT picture to relate $\mathcal{Q}^A$ and $\mathcal{Q}^A$. $\mathcal{Q}^L$ acts trivially on Ψ_R and vice-versa for $\mathcal{Q}^R$ and Ψ_L , so we can immediately calculate the charges q_A and q_V for Ψ_L and Ψ_R like so:

$$\begin{aligned} [\mathcal{Q}^V, \Psi_R^\dagger] &= [\mathcal{Q}^R, \Psi_R^\dagger] = +\Psi_R^\dagger, & [\mathcal{Q}^V, \Psi_L^\dagger] &= [\mathcal{Q}^L, \Psi_L^\dagger] = +\Psi_L^\dagger, \\ [\mathcal{Q}^A, \Psi_R^\dagger] &= [\mathcal{Q}^R, \Psi_R^\dagger] = +\Psi_R^\dagger, & [\mathcal{Q}^A, \Psi_L^\dagger] &= [-\mathcal{Q}^L, \Psi_L^\dagger] = -\Psi_L^\dagger. \end{aligned} \quad (\clubsuit)$$

¹⁶We make frequent use of this word, so we clarify here that the reader should take “integrality” to mean “taking integer values”.

¹⁷We note that we interchange integrals in this derivation, and we do likewise later on. We will not be treating this type of analysis very carefully.

These charges $\mathcal{Q}^V$ and $\mathcal{Q}^A$ generate the symmetries $U(1)^V$ and $U(1)^A$ respectively. That S is invariant under the action of these symmetries follows just as it did for $U(1)^R$ and $U(1)^L$ above, and we won't show it here. Because Ψ_L , Ψ_R and their conjugates have well-defined charges under $U(1)^V$ and $U(1)^A$ they also live in simultaneous charge sectors of the symmetries and as such we expect the symmetries to commute, and in fact they do. Expanding definitions we find:

$$e^{i\phi_1 \mathcal{Q}^V} e^{i\phi_2 \mathcal{Q}^A} = e^{i\phi_1(\mathcal{Q}^R + \mathcal{Q}^L)} e^{i\phi_2(\mathcal{Q}^R - \mathcal{Q}^L)},$$

whence the commutation of $\mathcal{Q}^R$ and $\mathcal{Q}^L$ allows us to treat these exponents naïvely, yielding:

$$e^{i\phi_2(\mathcal{Q}^R - \mathcal{Q}^L)} e^{i\phi_1(\mathcal{Q}^R + \mathcal{Q}^L)} = e^{i\phi_2 \mathcal{Q}^A} e^{i\phi_1 \mathcal{Q}^V}.$$

The connected component of our symmetry group is $[U(1)^V \times U(1)^A] / \mathbb{Z}_2$, where the quotient is taken due to $\mathcal{Q}^A$ and $\mathcal{Q}^V$ both acting the same for $\phi_1 = \phi_2 = \pi$. The proof is provided in [appendix C](#).

3.2 On the lattice

Here we will be writing down a Hamiltonian theory, so we must begin by writing our above continuum theory in the Hamiltonian formalism. Our Lagrangian density is given by $\mathcal{L} = i\bar{\Psi}\Gamma^\mu\partial_\mu\Psi$, and hence our canonical momenta by $\Pi_\Psi = \frac{\partial\mathcal{L}}{\partial(\partial_t\Psi)} = i\bar{\Psi}\Gamma^0 = i\Psi^\dagger$ and $\Pi_{\Psi^\dagger} = \frac{\partial\mathcal{L}}{\partial(\partial_t\Psi^\dagger)} = 0$. Simply performing our Legendre transform we arrive at:

$$\begin{aligned} H &= \int dx \mathcal{H} = \int dx \left(\Pi_\Psi \partial_t \Psi + \Pi_{\Psi^\dagger} \partial_t \Psi^\dagger - \mathcal{L} \right) = \int dx \left(i\Psi^\dagger \partial_t \Psi - i\bar{\Psi}\Gamma^0 \partial_t \Psi - i\bar{\Psi}\Gamma^1 \partial_x \Psi \right) \\ &= -i \int dx \bar{\Psi}\Gamma^1 \partial_x \Psi = -i \int dx \left(\Psi_R^\dagger \partial_x \Psi_R - \Psi_L^\dagger \partial_x \Psi_L \right) =: H_R + H_L. \end{aligned}$$

It is this theory that we will be putting on the lattice. Typically, when writing the kinetic term of a scalar lattice model, we may simply employ a hopping term $\hat{T}_\mu + \text{h.c.}$ ¹⁸. This term is isotropic, and as such commutes with the flip charge operator $\hat{F}$, which is the lattice version of the spatial parity operator P (exhaustive details in [appendix B](#)). It would be reasonable to figure that a free theory is symmetric under parity, but *au contraire*, the continuum theory is not!:

$$\begin{aligned} H_R &= -i \int dx \Psi_R(x)^\dagger \partial_x \Psi_R(x) \xrightarrow{P} -i \int dx \Psi_R^\dagger(-x) \partial_x \Psi_R(-x) \\ &= -i \int dx' \Psi_R^\dagger(x') (-\partial_{x'}) \Psi_R(x') = +i \int dx \Psi_R^\dagger(x) \partial_x \Psi_R(x). \end{aligned}$$

This changes the sign of our Hamiltonian, interchanging left-movers and right-movers. So, in our lattice Hamiltonian, we want our flip parity to act as $\text{Ad}_{e^{i\frac{\pi}{2}\hat{F}}}(\hat{H}) = -\hat{H}$ by analogy. The theory that respects this symmetry constraint is made with $i\hat{T}_\mu + \text{h.c.} = i\hat{T}_\mu - i\hat{T}_{-\mu}$:

$$\hat{H} = -i \sum_{j=1}^L \left(c_j^\dagger c_{j+1} + c_j c_{j+1}^\dagger \right) = i \sum_{j=1}^L c_{j+1}^\dagger c_j + \text{h.c.}$$

Of course, $\hat{H}$ resembles the kinetic term $\hat{T}_\mu + \text{h.c.}$. This imaginary prefactor is what makes our theory

¹⁸We use h.c. as shorthand for adding the hermitian conjugate of the other component.

anisotropic and, in particular, chiral. Let's check how H behaves under flip parity:

$$\begin{aligned}
\exp\left(i\frac{\pi}{2}\text{ad}_{\hat{F}}\right)\hat{H} &= i\sum_{j=1}^L\left(e^{i\frac{\pi}{2}\hat{F}}c_{j+1}^\dagger c_j e^{-i\frac{\pi}{2}\hat{F}}\right) + \text{h.c.} = i\sum_{j=1}^L\left(e^{i\frac{\pi}{2}\hat{F}}c_{j+1}^\dagger e^{i\frac{\pi}{2}\hat{F}}e^{-i\frac{\pi}{2}\hat{F}}c_j e^{-i\frac{\pi}{2}\hat{F}}\right) + \text{h.c.} \\
&= i\sum_{j=1}^L c_{L-(j+1)}^\dagger c_{L-j} + \text{h.c.} = i\sum_{j'=0}^{L-1} c_{j'}^\dagger c_{j'+1} + \text{h.c.} = \text{h.c.} - i\sum_{j'=0}^{L-1} c_{j'+1}^\dagger c_{j'} \\
&= \text{h.c.} - i\sum_{j=1}^L c_{j+1}^\dagger c_j = -\hat{H},
\end{aligned}$$

which is exactly what we expected.

Remark 3.2. We were able to change the sum above from $\sum_{j'=0}^{L-1} c_{j'+1}^\dagger c_{j'}$ to $\sum_{j=1}^L c_{j+1}^\dagger c_j$ only because $c_{0+1}^\dagger c_0 = (-1)^{2\nu} c_{L+1}^\dagger c_L = c_{L+1}^\dagger c_L$. Because we have fermion bilinears, we are free from having to worry about the anti-periodic case. We will make use of this fact again in later calculations.

3.2.1 The vector charge

By construction through $\hat{T}_{\hat{\mu}}$, we find that this Hamiltonian conserves particle number and commutes with translation operators. It follows from this particle number conservation that:

$$Q^V := \sum_{j=1}^L \left(c_j^\dagger c_j - \frac{1}{2} \right) = \hat{n} - \frac{1}{2}L \quad (\diamond)$$

is conserved also. That this is a $U(1)$ charge is immediate from its form as a number operator. We draw the analogy with Q^V most clearly through the definition of the vector charge via the vector current $J_V^\mu = \bar{\Psi}\Gamma^\mu\Psi$ as:

$$Q^V = \int dx \bar{\Psi}\Gamma^0\Psi = \int dx \Psi^\dagger\Psi.$$

(See [Ton06].) This resembles, in structure, equation $(\diamond)$.¹⁹ c_i and $c_i^\dagger$ have well-defined vector charges:

$$[Q^V, c_i] = \sum_{j=1}^L [c_j^\dagger c_j, c_i] = \sum_{j=1}^L \left(c_j^\dagger c_j c_i - (\delta_{i-j}^\nu - c_j^\dagger c_i) c_j \right) = -c_i,$$

so c_i has vector charge -1 , and likewise $c_i^\dagger$ has vector charge $+1$. In [CPS25], they decompose the complex Dirac fermion c_j into real Majorana operators as $c_j = \frac{1}{2}(a_j + ib_j)$. Because a_j and b_j are real operators we find $a_j^\dagger = a_j$ and $b_j^\dagger = b_j$, whence $c_j^\dagger = \frac{1}{2}(a_j - ib_j)$.

Remark 3.3. Two important facts follow almost immediately from the identities $a_j = c_j + c_j^\dagger$ and $b_j = -i(c_j - c_j^\dagger)$. Firstly, that $\{a_j, a_{j'}\} = \{b_j, b_{j'}\} = 2\delta_{j-j'}^\nu$ and $\{a_j, b_{j'}\} = 0$. Secondly, we have the boundary conditions: $a_{j+L} = (-1)^\nu a_j$ and $b_{j+L} = (-1)^\nu b_j$.

¹⁹We dedicate a lot of time later in showing how Q^A flows to Q^A in the infinite-extent IR limit. A similar procedure follows for Q^V and $\bar{Q}^V$ but we take it as granted here.

Using these operators, we can decouple our Hamiltonian into a real and imaginary part:

$$\hat{H} = -\frac{i}{2} \sum_{j=1}^L (a_j a_{j+1} + b_j b_{j+1}) = \hat{H}_a + \hat{H}_b = \hat{H}_a \oplus \hat{H}_b.$$

With such a decomposition, we can separate our Fock space into “real” and “imaginary” subspaces; those on which $\hat{H}_a$ and $\hat{H}_b$ act respectively. These operators also allow us to recast our vector charge as a sum of on-site (and therefore ultra-local) charge density operators which bridge the real and imaginary parts of the Fock space:

$$Q^V = \sum_{j=1}^L \left(\frac{1}{4} (a_j - ib_j)(a_j + ib_j) - \frac{1}{2} \right) = \sum_{j=1}^L \left(\frac{1}{4} (1 + ia_j b_j - ib_j a_j + 1) - \frac{1}{2} \right) = \sum_{j=1}^L \frac{i}{2} a_j b_j =: \sum_{j=1}^L q_j^V.$$

It's easy to see that each of these charge density operators q_j^V can be simultaneously diagonalised by virtue of our above commutation relations for a_j and b_j ; q_j^V will commute right past q_i^V for $i \neq j$.

3.2.2 The axial charge

We now have the pleasure of introducing the axial charge as presented in the Chatterjee-Pace-Shao paper:

$$Q^A = \frac{1}{2} \sum_{j=1}^L (c_j + c_j^\dagger)(c_{j+1} - c_{j+1}^\dagger).$$

We accept this operator *ex machina* for now; although we will see the inspiration for this definition in section 4. Like the vector charge, this operator may be written in terms of ultra-local charge density operators $q_{j+\frac{1}{2}}^A$:

$$Q^A = \sum_{j=1}^L \frac{i}{2} a_j b_{j+1} =: \sum_{j=1}^L q_{j+\frac{1}{2}}^A.$$

This charge density description will allow us to promote it to a gauge symmetry later. For the time being, however, the onus is on us to demonstrate that Q^A actually generates a $U(1)$ symmetry of our Hamiltonian. We begin by showing that $[Q^A, \hat{H}] = 0$:

$$\begin{aligned} [Q^A, \hat{H}] &= \frac{1}{4} \sum_{j,j'=1}^L [a_j b_j + 1, a_{j'} a_{j'+1} + b_{j'} b_{j'+1}] \\ &= \frac{1}{4} \sum_{j,j'=1}^L \left(2(\delta_{j-j'}^\nu a_{j'+1} - \delta_{j-j'-1}^\nu a_j') b_{j+1} + 2a_j (\delta_{j-j'+1}^\nu b_{j'+1} - \delta_{j-j'}^\nu b_{j'}) \right) \\ &= \frac{1}{2} \sum_{j=1}^L (a_{j+1} b_{j+1} - a_j b_{j+2} + a_{j-1} b_{j+1} - a_j b_j) = 0, \end{aligned}$$

where this last equality follows from our **fermion bilinear trick** from earlier. Next, we need to demonstrate that this is a $U(1)$ charge in particular, which will require integrality of the adjoint action.

Proposition 3.1. *We claim that $\text{spec}(Q^A) \subset \frac{1}{2}\mathbb{Z}$, which we will demonstrate by showing that the local charge density operators $q_{j+\frac{1}{2}}^A$ are simultaneously diagonalisable, and then by showing that each is an involution.*

Proof. Notice:

$$\begin{aligned}
[q_{j+\frac{1}{2}}^A, q_{j'+\frac{1}{2}}^A] &= q_{j+\frac{1}{2}}^A q_{j'+\frac{1}{2}}^A - q_{j'+\frac{1}{2}}^A q_{j+\frac{1}{2}}^A = -\frac{1}{4} (a_j b_{j+1} a_{j'} b_{j'+1} - a_{j'} b_{j'+1} a_j b_{j+1}) \\
&= -\frac{1}{4} (4\delta_{j-j'}^\nu \delta_{j-j'}^\nu - 2a_j a_{j'} \delta_{j-j'}^\nu - 2b_{j+1} b_{j'+1} \delta_{j-j'}^\nu) = -\delta_{j-j'} + \frac{1}{4} \delta_{j-j'}^\nu (2(a_j)^2 + 2(b_{j+1})^2) \\
&= -\delta_{j-j'} + \delta_{j-j'}^\nu = 0 \text{ for } 0 \leq |j - j'| < L.
\end{aligned}$$

As all the $q_{j+\frac{1}{2}}^A$'s commute, they may be simultaneously diagonalised. Now consider:

$$\left(2q_{j+\frac{1}{2}}^A\right)^2 = 4\frac{i^2}{4} a_j b_{j+1} a_j b_{j+1} = (a_j)^2 (b_{j+1})^2 = \mathbf{1},$$

so $2q_{j+\frac{1}{2}}^A$ is an involution and hence has eigenvalues ± 1 . Therefore $\text{spec}\left(q_{j+\frac{1}{2}}^A\right) = \{-\frac{1}{2}, +\frac{1}{2}\}$, hence:

$$\text{spec}\left(Q^A\right) = \text{spec}\left(\sum_{j=1}^L q_{j+\frac{1}{2}}^A\right) \subseteq \left\{-\frac{L}{2}, -\frac{L}{2} + 1, \dots, \frac{L}{2}\right\} \subseteq \frac{1}{2}\mathbb{Z}.$$

□

The spectrum of the adjoint action ad_{Q^A} is given by differences in the eigenvalues of Q^A , and so $\text{span}(\text{ad}_{Q^A}) \subset \mathbb{Z}$, from which we are guaranteed that we have a well-defined $U(1)^A$ action. The Nielsen-Ninomiya theorem allows us to know *a priori* one of the following facts:

- (a) $[Q^V, Q^A] \neq 0$, or
- (b) The charge sectors of Q^A do not capture left- and right-movers.

Why is this? Suppose, for the sake of contradiction, that Q^V commuted with Q^A , and therefore had simultaneous charge sectors in the Hilbert space. Restricting to a charge sector where, for example, $q_V = q_A$, we must have *only* left-movers in our theory, violating the Nielsen-Ninomiya theorem. The two options above are not mutually exclusive, so showing that (a) is true does not imply that (b) is false. All the same:

Proposition 3.2. $[Q^V, Q^A] \neq 0$.

Proof.

$$\begin{aligned}
[Q^V, Q^A] &= \sum_{j,j'=1}^L \left[\frac{i}{2} a_j b_j, \frac{i}{2} a_{j'} b_{j'+1} \right] = -\frac{1}{4} \sum_{j,j'=1}^L (a_j b_j a_{j'} b_{j'+1} - a_{j'} b_{j'+1} a_j b_j) \\
&= -\frac{1}{4} \sum_{j,j'=1}^L (\delta_{j-j'}^\nu \delta_{j-j'-1}^\nu - 2\delta_{j-j'-1}^\nu a_j a_{j'} - 2\delta_{j-j}^\nu b_j b_{j'+1}) \\
&= -\frac{1}{2} \sum_{j=1}^L (a_j a_{j+1} - b_j b_{j+1}) \neq 0.
\end{aligned}$$

□

This is, at the very least, what we expect from a lattice analogue to the axial charge, but we will go on in the next section to show the exact correspondence between Q^A and $\mathcal{Q}^A$. The non-vanishing of this commutator is the beginning of a much deeper story we will explore later in section 5, but it suffices to note for now that, in the continuum limit, $[Q^V, Q^A]$ goes to zero.²⁰

²⁰This is shown in the appendix of [CPS25] and holds in particular within the sense of vanishing matrix elements.

4 Resolving the continuum in the IR

We will finally get into the business of finding the continuum theory as a low-energy effective field theory of the infinite-extent lattice. In order to do this, we need to identify the spectrum of the theory, which will involve grappling with the momentum picture of our theory. We diagonalise $\hat{H}$ by diagonalising $\hat{H}_a$ and $\hat{H}_b$ separately. Focusing on $\hat{H}_b$, we define crystal momentum modes:

$$\beta_k := \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{-2\pi i j k / L} b_j.$$

We see immediately that $\beta_k^\dagger = \beta_{-k}$. The allowed values of k depend on the boundary conditions, as a shift by L should leave β_k invariant, hence:

$$\beta_k = \beta'_k = \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{-2\pi i (j+L)k / L} b_{j+L} = e^{-2\pi i k} (-1)^\nu \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{-2\pi i j k / L} b_j \implies e^{-2\pi i k} = (-1)^\nu.$$

$$\therefore k \in \begin{cases} \mathbb{Z} & \nu = 0 \\ \mathbb{Z} + \frac{1}{2} & \nu = 1 \end{cases}, \text{ or simply } k \in \mathbb{Z} + \frac{\nu}{2}.$$

Remark 4.1. Recall the Brillouin zone of our theory is the interval $(0, L]$. The classic implication of this within the context of Bloch's theorem is that we can only resolve the physical (mechanical) momentum of any given particle modulo L . This lends a different perspective to our process of finding a continuum EFT from our theory. We can reinterpret our process of taking L large and looking at (relatively) smaller frequencies as being able to take L larger than any physical momentum we would like to discuss such that our crystal momenta and physical momenta become equal.

Proposition 4.1. *We claim that, up to a constant, $\hat{H}_b = \frac{1}{2} \sum_{0 \leq k < L} \sin(2\pi k / L) \beta_k^\dagger \beta_k$.*

First, we need to establish the commutation relations of our Fourier operators, starting with the Fourier orthogonality formulae. For $\Delta k \in \mathbb{Z}$, we can simply observe that $e^{2\pi i \Delta k / L}$ is an L -th root of unity and hence:

$$\left(e^{2\pi i \Delta k / L} - 1 \right) \left(\sum_{j=1}^L e^{2\pi i j \Delta k / L} \right) = e^{2\pi i \Delta k} - 1 = 0.$$

Hence $e^{2\pi i \Delta k / L} + \dots + e^{2\pi i \Delta k (L-1) / L} + 1 = \begin{cases} L & \Delta k = 0 \pmod{L} \\ 0 & \text{else} \end{cases} = L \delta_{\Delta k}$, where we're once again using

our periodic delta functions. For the other Fourier orthogonality equation, we will need to split the momenta into their integral and fractal parts. Here, $j \in \mathbb{Z}$:

$$\sum_{0 \leq k < L} e^{2\pi i j k / L} = \sum_{0 \leq k < L} e^{2\pi i j [k] / L} e^{2\pi i j \{k\} / L} = e^{\nu \pi i j / L} \sum_{[k]=0}^{L-1} e^{2\pi i j [k] / L} = e^{\nu \pi i j / L} \delta_j L = L \delta_j^\nu.$$

We see that the momentum orthogonality is dependant on the boundary conditions, which is only reasonable. The following holds for both $\nu = 0$ and $\nu = 1$ because in either case $k + k' \in \mathbb{Z}$:

$$\begin{aligned} \{\beta_k, \beta_{k'}\} &= \frac{1}{L} \sum_{j, j'=1}^L e^{-2\pi i (jk + j'k') / L} \{b_j, b_{j'}\} = \frac{2}{L} \sum_{j, j'=1}^L e^{-2\pi i (jk + j'k') / L} \delta_{j-j'} \\ &= \frac{2}{L} \sum_{j=1}^L e^{-2\pi i j (k+k') / L} = 2 \delta_{k+k'}, \end{aligned}$$

and so we can go on to our proof:

Proof.

$$\begin{aligned}
\frac{1}{2} \sum_{0 \leq k < L} \sin(2\pi k/L) \beta_k^\dagger \beta_k &= \frac{-i}{4L} \sum_{0 \leq k < L} \left(e^{2\pi i k/L} - e^{-2\pi i k/L} \right) \sum_{j,j'=1}^L e^{2\pi i j k} e^{-2\pi i j' k} b_j b_{j'} \\
&= \frac{-i}{4L} \sum_{j,j'=1}^L b_j b_{j'} \sum_{0 \leq k < L} \left(e^{2\pi i (1+j-j')k/L} - e^{2\pi i (-1+j-j')k/L} \right) \\
&= \frac{-i}{4L} \sum_{j,j'=1}^L b_j b_{j'} (\delta_{1+j-j'}^\nu - \delta_{-1+j-j'}^\nu) L \\
&= \frac{-i}{4} \sum_{j=1}^L (b_j b_{j+1} - b_{j+1} b_j) = -\frac{i}{2} \sum_{j=1}^L b_j b_{j+1} = \hat{H}_b
\end{aligned}$$

□

4.1 Fermi sea picture

Naïvely, it looks like $\hat{H}_b$ has dispersion $E_{\text{naïve}}(k) = \frac{1}{2} \sin(2\pi k/L)$. This isn't quite right, as we have something of a normal ordering ambiguity here; the field creation operators for states with $k > L/2$ are also field annihilation operators for states with $k < L/2$: $\beta_{L-k}^\dagger \beta_{L-k} = \beta_{-k}^\dagger \beta_{-k} = \beta_k \beta_k^\dagger$. For $0 \leq k \leq L/2$, we can interpret the action of $\beta_k^\dagger$ on the ground state $|\Omega\rangle$ as the simultaneous creation of a particle with energy $\frac{1}{2} \sin(2\pi k/L) \geq 0$ and annihilation of a particle with energy $\frac{1}{2} \sin(-2\pi k/L) \leq 0$, increasing the overall energy by an amount $\sin(2\pi k/L)$. This interpretation only works in the Fermi sea picture, where our ground state $|\Omega\rangle$ is given by:

$$|\Omega\rangle := \left(\prod_{L/2 < k < L} \beta_k^\dagger \right) |\text{vac}\rangle,$$

where $|\text{vac}\rangle$ is the free vacuum with no particles. This is the arrangement where we've filled all the negative

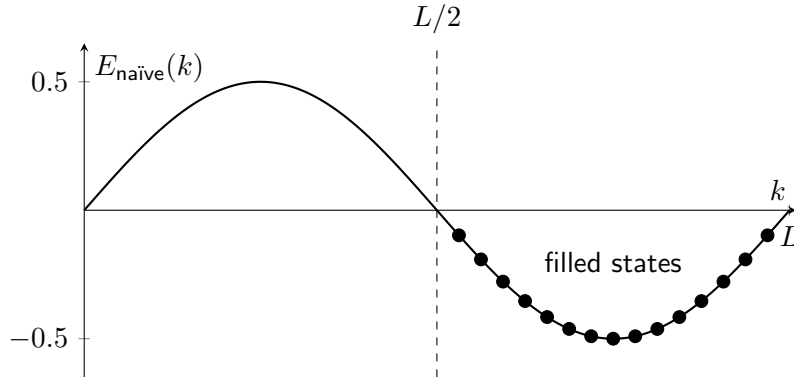


Figure 8: Fermi sea for the naïve lattice dispersion relation.

energy states and cannot get any more free energy by removing particles from our setup. In the case where $\nu = 0$, this ground state is part of a degenerate quartet, as we are free to add particles at the Fermi points $k = 0$ and $k = L/2$ where $\sin(2\pi k/L) = 0$, resulting in the ground states $\{|\Omega\rangle, \beta_0^\dagger |\Omega\rangle, \beta_{L/2}^\dagger |\Omega\rangle, \beta_{L/2}^\dagger \beta_0^\dagger |\Omega\rangle\}$. For $\nu = 1$ the Fermi points are bad quantum numbers and we have a unique ground state.

We'll now explicitly write the dispersion relation for our Hamiltonian:

$$\begin{aligned}
\frac{1}{2} \sum_{0 \leq k < L} \sin(2\pi k/L) \beta_k^\dagger \beta_k &= \frac{1}{2} \left(\sum_{0 < k < L/2} + \sum_{L/2 < k < L} \right) \sin(2\pi k/L) \beta_k^\dagger \beta_k \\
&= \frac{1}{2} \sum_{0 < k < L/2} \left(\sin(2\pi k/L) \beta_k^\dagger \beta_k + \sin(-2\pi k/L) \beta_{-k}^\dagger \beta_{-k} \right) \\
&= \frac{1}{2} \sum_{0 < k < L/2} \sin(2\pi k/L) \left(\beta_k^\dagger \beta_k - \beta_k \beta_k^\dagger \right) \\
&= \left(\sum_{0 < k < L/2} \sin(2\pi k/L) \beta_k^\dagger \beta_k \right) - \left(\sum_{0 < k < L/2} \sin(2\pi k/L) \right).
\end{aligned}$$

We see that we have picked up an extra constant term $-\sum_{0 < k < L/2} \sin(2\pi k/L)$. This is precisely the ground state energy of our filled Fermi sea: a contribution of $-|E(k)|$ from each negative energy state. We can put this v.e.v. in a tidy closed form like so: Notice that $\sum_{0 < k < L/2} \sin(2\pi k/L) = \text{Im} \left(\sum_{0 < k < L/2} \omega^k \right)$ where ω is the principal L -th root of unity. For $\nu = 0$, this becomes simply $\text{Im}(\omega^1 + \dots + \omega^{\frac{L}{2}-1}) = \text{Im} \left((\omega^{L/2} - \omega)/(\omega - 1) \right) = \text{Im} \left((1 + \omega)/(1 - \omega) \right) = \text{Im}((\omega - \bar{\omega})/(2 - \omega - \bar{\omega})) = \sin(2\pi/L)/(1 - \cos(2\pi/L)) = \cot(\pi/L)$. For $\nu = 1$, the sum is only slightly modified, yielding $\text{Im}(\omega^{\frac{1}{2}}(1 + \omega + \dots + \omega^{\frac{L}{2}-1}))$. We recognise this as the addition of 1 to $\omega^1 + \dots + \omega^{\frac{L}{2}-1}$ composed with a rotation to make it purely imaginary again, hence this simplifies to $\sqrt{1 + \cot^2(\pi/L)} = \csc(\pi/L)$.

$$\therefore \hat{H}_b = \sum_{0 < k < L/2} \sin(2\pi k/L) \beta_k^\dagger \beta_k - \begin{cases} \cot(\pi/L) & \nu = 0 \\ \csc(\pi/L) & \nu = 1 \end{cases}.$$

Remark 4.2. We notice that our constant terms both diverge as $L \rightarrow \infty$. This is to be expected from our Fermi sea picture, but could still be troubling. However all we need notice is that $\cot(x\pi)$ has only a simple pole at $x = 0$ and hence $x \cot(x\pi)$ has a well defined limit of $\frac{1}{\pi}$ as $x \rightarrow 0$. For this reason, the energy density $\frac{1}{L} \cot(\pi/L)$ has a well-defined limit as $L \rightarrow \infty$. The same holds for $\frac{1}{L} \csc(\pi/L)$ and hence in either case our Hamiltonian density $\mathcal{H}$ is well defined in the limit.

4.2 The low-energy effective field theory

It is at this point that we can identify fermion doublers in our system. The spectrum of our theory is given by $E(k) = \sin(2\pi k/L)$ for $k \in [0, L/2]$. Notice that we have a closed interval as the opposite ends of this interval are *not* to be identified, which has the additional effect of equipping our effective momentum space with a trivial fundamental group. In the IR limit as $\Gamma \rightarrow 0$, we are left only with neighbourhoods around our two Fermi points at $k = 0$ and $k = L/2$. Taylor expanding around each of these points yields the following linear approximations: $E(L\varepsilon) = 2\pi\varepsilon + \mathcal{O}(\varepsilon^2)$ and $E(L/2 + L\varepsilon) = -2\pi\varepsilon + \mathcal{O}(\varepsilon^2)$. Just for now, imagining that these ε terms encapsulate continuous momenta, we have identified left-movers ($\frac{\partial E}{\partial \varepsilon} < 0$) and right-movers ($\frac{\partial E}{\partial \varepsilon} > 0$) in our spectrum.²¹

Remark 4.3. It is at this point that we see the manifestation of the Nielsen-Ninomiya theorem for our theory: the positive-slope node in our spectrum is matched up with a negative-slope node in our spectrum, and these correspond directly to right- and left-movers respectively. The fact that each positive slope is always matched with a negative slope forms the topological discussion of the theorem as presented in [NN81].

²¹The authors of [CPS25] use the opposite convention for left- and right-movers.

We now lay out a more constructive probe of this phenomenon, beginning by defining two spatial field operators:

$$\chi_R(x) := \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i x(q/L)} \beta_q \quad \text{and} \quad \chi_L(x) := \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i x(q/L)} \beta_{\frac{L}{2}-q}.$$

Note that x here is not fixed to be integral, so we can discuss the continuum between lattice sites and the aliasing phenomena appearing therein. The spatial extent of this x is taken to be the finite-volume interval $[-V/2, V/2]$, where we will be taking V to ∞ afterwards.²² While these are spatial field operators, they differ from the b_j operators in many ways other than non-integrality. The momentum cut-off comes from separation of scales in momentum units:

$$1 \ll 2\pi q \leq L\Gamma \ll L.$$

The imposition of the $0 \leq q < \frac{L}{2\pi}\Gamma$ inequality essentially implements a band-pass filter²³ on our b_j operators. We do note that a sharp cut-off is not the only valid prescription here, and it has associated issues as we will see. Because these spatial operators are limited in their momentum extent, they are not fully spatially resolved; these operators have messy commutation relations as the orthogonality is smeared in the band-pass filtering. The commutation relation is given by:

$$\{\chi_R(x), \chi_R(y)\} = \frac{1}{V} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i q(x-y)/L}.$$

In the $L \rightarrow \infty$ limit, this expression converges to $\frac{\Gamma L}{\pi V} \text{sinc}(\Gamma(x-y))$.²⁴ Our goal is to find a continuum field theory, so let's define the Hamiltonian $\hat{H}_\chi$ to take the form of the free continuum massless fermion Hamiltonian:

$$\hat{H}_\chi := -i \int dx \left(\chi_R^\dagger \partial_x \chi_R - \chi_L^\dagger \partial_x \chi_L \right),$$

where the integral is over $[-V/2, V/2]$. We want to connect this with our actual theory in the IR infinite-extent limit, and so we unpack our definitions of the χ fields.

$$\hat{H}_\chi = \frac{2\pi}{VL} \sum_{0 \leq p, q < \frac{L}{2\pi}\Gamma} \left(q \beta_p^\dagger \beta_q \left[\int dx e^{-2\pi i(p-q)x/L} \right] + p \beta_{\frac{L}{2}-p}^\dagger \beta_{\frac{L}{2}-q} \left[\int dx e^{2\pi i(p-q)x/L} \right] \right).$$

In the limit as $V \rightarrow \infty$, the expression $\frac{1}{V} \int dx e^{-2\pi i(p-q)x/L} \rightarrow \delta_{p-q}$, and so our expression simplifies to:

$$\hat{H}_\chi \xrightarrow{V \rightarrow \infty} \sum_p \left(\frac{2\pi p}{L} \beta_p^\dagger \beta_p + \frac{2\pi p}{L} \beta_{\frac{L}{2}-p}^\dagger \beta_{\frac{L}{2}-p} \right).$$

We see the emergence of the linear approximation solutions in this Hamiltonian. The spectrum itself is not surprising, after all, we have essentially written down a Hamiltonian for two free fermion fields with momentum $P = \frac{2\pi q}{L}$, and we have found that each has energy given by $|E| = |P|$. The fact to write home about is the more subtle observation that we have an explicit identification between a continuum field theory

²²We deal with V instead of taking the extent of our theory to be $(0, L]$ here. That this is allowed is immediate (we are constructing the fields) but the reason we are doing this is because we need to be careful about IR volume divergences here.

²³Once again permitting ourselves terminology from the world of sound processing.

²⁴Standard result of Dirichlet kernels. See [BBT97, Chapter 15.2] and imagine taking $n \rightarrow \infty$ for $D_n(t)$.

and the IR lattice theory in the large L approximation; we have explicitly constructed our low-energy EFT. We will use this Hamiltonian to construct a continuous momentum operator around the Fermi points in order to demonstrate the construction of Q^A in [CPS25]. Using the standard definition of the momentum operator, we find:

$$\hat{P}_\chi = -i \int dx \left(\chi_R^\dagger \partial_x \chi_R + \chi_L^\dagger \partial_x \chi_L \right) \xrightarrow{V \rightarrow \infty} \sum_p \left(\frac{2\pi p}{L} \beta_p^\dagger \beta_p - \frac{2\pi p}{L} \beta_{\frac{L}{2}-p}^\dagger \beta_{\frac{L}{2}-p} \right),$$

where we've performed the same treatment as for $\hat{H}_\chi$.²⁵ It's very simple to check that $\hat{P}_\chi \beta_p^\dagger |\Omega\rangle = \frac{2\pi p}{L} \beta_p^\dagger |\Omega\rangle$ and $\hat{P}_\chi \beta_{\frac{L}{2}-p}^\dagger |\Omega\rangle = -\frac{2\pi p}{L} \beta_{\frac{L}{2}-p}^\dagger |\Omega\rangle$, which tells us that this momentum operator has, as its eigenvalues, the *physical continuous* momenta. So, we do not know a priori²⁶ how this operator behaves when we try to “generate translations” using it. In the continuum, momentum famously generates translation, and a similar relation exists on the lattice between crystal momentum generating lattice translations, but we do *not* expect to find that $\hat{P}_\chi$ generates lattice translations as such. So what exactly is the relationship between $\hat{P}_\chi$ and $\hat{T}_b$?²⁷

4.3 Matching Q^A to $\mathcal{Q}^A$

Proposition 4.2. *In the IR theory, $\hat{T}_b = \mathcal{C}^L \hat{T}_\chi$, where $\hat{T}_\chi := \exp(-i\hat{P}_\chi)$.*

Proof. We can repeat all of the same arguments above for the real Majorana operators a_j where we define:

$$\lambda_R(x) := \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i q x / L} \alpha_q \text{ and } \lambda_L(x) := \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i q x / L} \alpha_{\frac{L}{2}-q}, \text{ for } \alpha_k := \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{-2\pi i j k / L} a_j.$$

We can then identify the Dirac right- and left-movers of the continuum IR theory as $\psi_R := \lambda_R + i\chi_R$ and $\psi_L := \lambda_L + i\chi_L$. Recalling that the charge conjugation operator $\mathcal{C}^L$ acts by sending ψ_L to $\psi_L^\dagger$ (and vice versa), we must have that χ_L maps to $-\chi_L$.

To show the equivalence of the operators, we simply compare the action of the fields $|\chi_R(x)\rangle := \chi_R^\dagger(x) |\Omega\rangle$ and $|\chi_L(x)\rangle := \chi_L^\dagger(x) |\Omega\rangle$ under the operators $\hat{T}_b$ and $\mathcal{C}^L \hat{T}_\chi$. We begin with the details of $\chi_R(x)$:

$$\hat{T}_b |\chi_R(x)\rangle = \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i x q / L} \hat{T}_b \beta_q^\dagger |\Omega\rangle = \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i (x+1) q / L} \beta_q^\dagger |\Omega\rangle = |\chi_R(x+1)\rangle,$$

$$\mathcal{C}^L \hat{T}_\chi |\chi_R(x)\rangle = \frac{\mathcal{C}^L}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i x q / L} e^{-i\hat{P}_\chi} \beta_q^\dagger |\Omega\rangle = \frac{\mathcal{C}^L}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i (x+1) q / L} \beta_q^\dagger |\Omega\rangle = \mathcal{C}^L |\chi_R(x+1)\rangle.$$

The *left-moving* charge conjugation operator doesn't care about χ_R , and so we find that $\hat{T}_b |\chi_R(x)\rangle = \mathcal{C}^L \hat{T}_\chi |\chi_R(x)\rangle$. For the χ_L case, we find:

$$\hat{T}_b |\chi_L(x)\rangle = \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i (x+1) q / L} e^{-2\pi i (L/2) q / L} \beta_{\frac{L}{2}-q}^\dagger |\Omega\rangle = -|\chi_L(x+1)\rangle,$$

$$\mathcal{C}^L \hat{T}_\chi |\chi_L(x)\rangle = \frac{\mathcal{C}^L}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i x q / L} e^{-i\hat{P}_\chi} \beta_{\frac{L}{2}-q}^\dagger |\Omega\rangle = \mathcal{C}^L |\chi_L(x+1)\rangle = -|\chi_L(x+1)\rangle.$$

□

²⁵In what follows, we simply take this $V \rightarrow \infty$ limit as implied. We only care about V as an IR regulator, and we won't want to deal with it any more.

²⁶This author did not know, notwithstanding that the reader may be very bright.

²⁷This is the 1-site hopping term on *only* the b field. i.e., $\hat{T}_b b_j \hat{T}_b^{-1} = b_{j+1}$.

Remark 4.4. In the above proof, we make repeated use of the simple relation that $\hat{T}_b \beta_k |\Omega\rangle = e^{2\pi i k/L} \beta_k |\Omega\rangle$ and $\hat{T}_b \beta_k^\dagger |\Omega\rangle = e^{-2\pi i k/L} \beta_k^\dagger |\Omega\rangle$. These follow almost immediately from the definition of β_k and $\hat{T}_b$.

What's more, for a fixed low-energy momentum mode, the $\hat{T}_\chi$ term goes to unity in the infinite-extent limit (simply because $\hat{P}_\chi \sim 2\pi p/L \rightarrow 0$ for $L \rightarrow \infty$ and p fixed) and so we identify $\mathcal{C}^L$ in our low-energy theory with $\hat{T}_b$ in the lattice theory. This relation, combined with our observation (♠) from earlier, is the construction that the Chatterjee-Pace-Shao paper uses to define Q^A , and it should obey

$$Q^A = \hat{T}_b Q^V (\hat{T}_b)^{-1}.$$

Plugging in definitions, we find exactly the same:

$$\hat{T}_b Q^V (\hat{T}_b)^{-1} = \frac{i}{2} \sum_j \hat{T}_b (a_j b_j) (\hat{T}_b)^{-1} = \frac{i}{2} \sum_j a_j b_{j+1} = Q^A.$$

In order to discuss the behaviour of these ψ_L, ψ_R operators, we use the following notation from [CPS25]: $\gamma_k := \frac{1}{2}(\alpha_k + i\beta_k)$, from which it quickly follows that:

$$\psi_R(x) = \frac{2}{\sqrt{L}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{2\pi i q x/L} \gamma_q \quad \text{and} \quad \psi_L(x) = \frac{2}{\sqrt{L}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} e^{-2\pi i q x/L} \gamma_{\frac{L}{2}-q}.$$

We are now positioned to be able to show the analytical connection between our Q^A operator and the continuum $\mathcal{Q}^A$ axial charge. We need an exact relation between Q^A and our crystal momentum modes in order to discuss the infinite-extent lattice with $L \rightarrow \infty$:

Proposition 4.3. $[Q^A, \gamma_k^\dagger] = \cos \frac{2\pi k}{L} \gamma_k^\dagger + i \sin \frac{2\pi k}{L} \gamma_{-k}$.

Proof. Simply expanding definitions yields:

$$[Q^A, \gamma_k^\dagger] = \frac{i}{4} \sum_{j=1}^L [a_j b_{j+1}, \alpha_k^\dagger - i\beta_k^\dagger].$$

We will deal with the $\alpha_k^\dagger$ component, and it should be clear how the $\beta_k^\dagger$ changes in the same way. Expanding:

$$[a_j b_{j+1}, \alpha_k^\dagger] = \frac{1}{\sqrt{L}} \sum_{l=1}^L e^{2\pi i l k/L} [a_j b_{j+1}, a_l] = -\frac{1}{\sqrt{L}} \sum_{l=1}^L e^{2\pi i l k/L} \{a_j, a_l\} b_{j+1}.$$

Using our commutation relation, and expanding the periodic delta function which emerges, we find:

$$[a_j b_{j+1}, \alpha_k^\dagger] = -\frac{2}{\sqrt{L}} e^{2\pi i j k/L} b_{j+1} \quad \text{and likewise,} \quad [a_j b_{j+1}, \beta_k^\dagger] = \frac{2}{\sqrt{L}} e^{2\pi i (j+1)k/L} a_j.$$

Plugging these equations into the first commutator equation, and making use of the BCs, we find:

$$\begin{aligned} [Q^A, \gamma_k^\dagger] &= -\frac{i}{2} \left(e^{-2\pi i k/L} \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{2\pi i j k/L} b_j + i e^{2\pi i k/L} \frac{1}{\sqrt{L}} \sum_{j=1}^L e^{2\pi i j k/L} a_j \right) \\ &= \frac{1}{2} \left(e^{2\pi i k/L} \alpha_k^\dagger - i e^{-2\pi i k/L} \beta_k^\dagger \right) = \frac{1}{2} \left(\alpha_k^\dagger - i\beta_k^\dagger \right) \cos(2\pi k/L) + \frac{1}{2} \left(\alpha_k^\dagger + i\beta_k^\dagger \right) i \sin(2\pi k/L) \\ &= \cos \left(\frac{2\pi k}{L} \right) \gamma_k^\dagger + i \sin \left(\frac{2\pi k}{L} \right) \gamma_{-k}, \end{aligned}$$

where the identification of γ_{-k} with $\frac{1}{2}(\alpha_k^\dagger + i\beta_k^\dagger)$ is just straightforwardly looking at the definitions of each side. $\square$

We are able to use linear approximations of sine and cosine to see how the above relation looks in the infinite-extent lattice. [CPS25]'s analysis follows as:

$$\lim_{L \rightarrow \infty} [Q^A, \gamma_q^\dagger] = (1)\gamma_q^\dagger + (0)\gamma_{-q} = \gamma_q^\dagger, \quad \lim_{L \rightarrow \infty} [Q^A, \gamma_{\frac{L}{2}-q}^\dagger] = (-1)\gamma_{\frac{L}{2}-q}^\dagger + (0)\gamma_{-\frac{L}{2}+q} = -\gamma_{\frac{L}{2}-q}^\dagger,$$

for a fixed value of q . This discussion is more heuristic than complete, as in the limit of $L \rightarrow \infty$, the operator $\gamma_q^\dagger$ isn't quite defined; because (a), $\gamma_q^\dagger$ is an operator which acts on a finite-dimensional Hilbert space whose dimensionality is explicitly dependent on L , and (b), because L appears in the definition of $\gamma_q^\dagger$ in a way that makes it unclear what γ_q would be in the infinite-extent lattice. We propose a view of this limit within the context of the operator norm $\|\cdot\|_L$. Note that, for each value of L , this norm is different.

Proposition 4.4. $\lim_{L \rightarrow \infty} \|[Q^A, \gamma_q^\dagger] - \gamma_q^\dagger\|_L = 0$.

Proof. Below, we make use of the fact that, for a fermionic operator f the operator norm $\|f\|_L = \sup_{\|\psi\|=1} \|f|\psi\rangle\|$ is equal to 1, which is saturated for $|\psi\rangle = |\text{vac}\rangle$.

$$\begin{aligned} \|[Q^A, \gamma_q^\dagger] - \gamma_q^\dagger\|_L &\leq \left| \cos \frac{2\pi q}{L} - 1 \right| \|\gamma_q^\dagger\|_L + \left| \sin \frac{2\pi q}{L} \right| \|\gamma_{-q}\|_L \\ &\leq \frac{1}{2} \left| \frac{2\pi q}{L} \right|^2 + \left| \frac{2\pi q}{L} \right| \sim \mathcal{O}\left(\frac{1}{L}\right). \end{aligned}$$

We can see then that the difference between $[Q^A, \gamma_q^\dagger]$ and $\gamma_q^\dagger$ goes to zero in the limit as $L \rightarrow \infty$. $\square$

It follows, *mutatis mutandis*, that $[Q^A, \gamma_{\frac{L}{2}-q}^\dagger]$ approaches $-\gamma_{\frac{L}{2}-q}^\dagger$ in the same sense. These two relations match the action that the axial charge should have on the right- and left-movers in the continuum as in equation (♣). We would like to make a tighter identification here, to show that the spatial wavefunctions $\psi_R(x)$ and $\psi_L(x)$ match up with $\Psi_R(x)$ and $\Psi_L(x)$ in the infinite-extent lattice in the IR. Note what occurs for our spatial fields under the same analysis as above:

$$\begin{aligned} \|[Q^A, \psi_R^\dagger(x)] - \psi_R^\dagger(x)\|_L &\leq \frac{2}{\sqrt{L}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} \left| e^{2\pi i q x/L} \right| \left| \left(\cos \frac{2\pi q}{L} - 1 \right) \gamma_q^\dagger - \left(\sin \frac{2\pi q}{L} \right) \gamma_{-q} \right|_L \\ &\leq \frac{2}{\sqrt{L}} \sum_{0 \leq q < \frac{L}{2\pi}\Gamma} \left(\frac{1}{2} \left| \frac{2\pi q}{L} \right|^2 + \left| \frac{2\pi q}{L} \right| \right) \\ &\leq \frac{1}{\sqrt{L}} \left(\frac{L}{\pi} \Gamma \right) \left(\frac{1}{2} \Gamma^2 + \Gamma \right) \sim \mathcal{O}\left(\Gamma^2 \sqrt{L}\right). \end{aligned}$$

This difference isn't as simple as the pure momentum modes, so there's no immediate analogue of the form $\lim_{L \rightarrow \infty} [Q^A, \psi_R^\dagger] = \psi_R^\dagger$. Instead, we must simultaneously decrease our IR cut-off in such a way that every sixteen-fold increase in L must incur at least a halving of Γ . This careful balancing of restricting to a more infra-red spectrum is a little awkward and can be dealt with in a few other ways beyond this band-pass filtering. If the low-momentum modes were selected with a smeared cut-off instead of $0 \leq q < \frac{L}{2\pi}\Gamma$, possibly convolving our step function with some smooth bump function, we could have an easier restriction on the rate at which Γ must be decreased with increasing L . The other, far simpler option is to have a momentum cut-off in lattice units, that is to say we have some fixed Γ_q independent of L such that $0 \leq q < \Gamma_q$ is our cut-off. This is less physically motivated than the IR cut-off Γ , but has the advantage of not needing to balance the two scale factors.

5 Lattice anomalies

We now turn our attention to how the chiral anomaly manifests on the lattice, and in particular how it relates to the Nielsen-Ninomiya theorem. We begin by discussing Hamiltonian lattice gauge theory *à la Susskind*, following [CPS25]'s gauging of the $U(1)^V$ and $U(1)^A$ symmetries. This construction will shed light on the obstruction to gauging both symmetries simultaneously, and we will finish with a discussion of how the non-commutativity of Q^V and Q^A matches the continuum chiral anomaly.

5.1 Gauging $U(1)^V$ and $U(1)^A$

We now come to our Rubicon, we must grapple with the details of gauging our symmetries on the lattice. Fortunately for us, this is a rich and beautiful theory; in this author's opinion moreso than the continuum version. This formulation was pioneered by Wilson in [Wil74], where he simultaneously (a) solidified the first proper theoretical understanding of *confinement* through his introduction of the Wilson loop and (b) laid the foundation for all of the most precise non-perturbative QCD calculations to come. We do not prove the results in this subsection, we simply describe how this process works for our symmetries.

Soon after, Banks, Kogut, and Susskind crafted the lattice Hamiltonian formulation of Wilson's theory in [KS75][BSK76], and it is this setup that we follow. The basic principle of the construction is that we have $U(1)$ gauge *group* operators which “live on the links” of the lattice. More precisely by this we mean that our site-to-site hopping terms are mediated by our $U(1)^V$ operators, which looks like the replacement $c_j c_{j+1} \mapsto c_j U_{j+\frac{1}{2}} c_{j+1}$ in the case of gauging the vector symmetry. In this way, as in the continuum, the gauge fields tell us how our internal degrees of freedom change as we roll around spacetime.²⁸

To build a $U(1)$ gauge theory we naturally have to introduce a new field, $A_{j+\frac{1}{2}}$, into our theory. From this field, each of these mediating $U(1)$ operators are constructed as unit-length *Wilson lines*.²⁹

$$U_{j+\frac{1}{2}} := \exp \left(i A_{j+\frac{1}{2}} \right).$$

Note that we work in the lattice equivalent of the Coulomb gauge such that we only deal with the spatial part of $A_{j+\frac{1}{2}}$ (there is no $A_{j+\frac{1}{2}}^0$ for us). We define $E_{j+\frac{1}{2}} := \frac{1}{2}(A_{j+\frac{3}{2}} - A_{j-\frac{1}{2}})$, the symmetric discrete derivative of A , and we enforce the discrete form of the canonical commutation relation:

$$[A_{j+\frac{1}{2}}, E_{l+\frac{1}{2}}] = i \delta_{j-l}^V.$$

With this interpretation, the Wilson lines represent electric flux lines on the lattice, and our process of gauging a symmetry involves connecting sites through these Wilson lines in direct analogy to minimal coupling.³⁰ This discussion is more complicated for the axial symmetry, because while the charge Q^A is ultra-local, it is not on-site, and so we don't want a $U(1)^A$ Wilson line from c_j to c_{j+1} . These are not eigen-operators of ad_{Q^A} and as such do not have a charge under the symmetry generated thereby. We must instead consider the change of basis field given by $c_j^A = \frac{1}{2}(a_j + i b_{j+1})$, and it is these operators on which Q^A acts on-site. i.e., we would be looking for a $U(1)^A$ connection between c_j^A and c_{j+1}^A . This basis change is actualised by conjugating our operators by $\hat{T}_b$:

$$c_j \mapsto c_j^A = \hat{T}_b c_j (\hat{T}_b)^{-1}, \text{ and } Q^V \mapsto Q^A = \hat{T}_b Q^V (\hat{T}_b)^{-1}.$$

²⁸We should be careful to note that our axial charge is not on-site, and so is not strictly an internal degree of freedom on the lattice.

²⁹In this work we take the coupling e to be 1 for convenience.

³⁰In some cases, such as that of the Peierls substitution, this analogy is exact, we replace momenta $\hat{p}$ with $\hat{p} - ie\hat{A}$, the effect of which is to add a phase between neighbouring sites.

From this perspective we can see that gauging Q^A is just a matter of gauging Q^V in a different basis.

The penultimate step in gauging on the lattice is the easiest, and simply involves adding an analogue to the pure gauge term $\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ to the Hamiltonian. The obvious correspondence is given by:

$$\frac{1}{2} \int dx E^2(x) \mapsto \frac{1}{2} \sum_{j=1}^L E_{j+\frac{1}{2}}^2,$$

where this first integral is what appears in the Schwinger model. From here, we can happily write down the Hamiltonian for the gauged vector symmetry as seen in [CPS25]:

$$\hat{H}_V = -i \sum_{j=1}^L \left(c_j^\dagger U_{j+\frac{1}{2}} c_{j+1} + c_j U_{j+\frac{1}{2}}^\dagger c_{j+1}^\dagger + \frac{i}{2} E_{j+\frac{1}{2}}^2 \right).$$

To find the Hamiltonian for our axial gauge, we simply conjugate our vector gauged Hamiltonian:

$$\hat{H}_A = \hat{T}_b \hat{H}_V (\hat{T}_b)^{-1} = -i \sum_{j=1}^L \left((c_j^A)^\dagger U_{j+\frac{1}{2}} c_{j+1}^A + c_j^A U_{j+\frac{1}{2}}^\dagger (c_{j+1}^A)^\dagger + \frac{i}{2} E_{j+\frac{1}{2}}^2 \right).$$

Remark 5.1. We see here the first reason that ultra-locality is an important property for a symmetry to be gauged: we needed to be able to change our basis to make our axial symmetry become on-site, through exchanging c_j for the eigen-operator of the ad_{Q^A} . This is possible only because Q^A itself was ultra-local, we could shift the effective lattice of b over by a fixed, finite amount to make up the difference. Without this property, we would not have the ability to gauge the symmetry.

The final stage of this process pertains to an analogue of the differential Gauss' law, how it generates gauge transformations, and how we use it to define the physical states of our theory [Ton18]. For the vector symmetry, the Gauss' law analogue in our theory presents as:

$$\left(E_{j+\frac{1}{2}} - E_{j-\frac{1}{2}} \right) |\psi\rangle = \left(q_j^V + \frac{(-1)^j}{2} \right) |\psi\rangle, \quad (\heartsuit)$$

analogous to $\partial_x E = \rho$.³¹ The strange $(-1)^j/2$ term is a result of the staggered fermion formalism, in which the combination on the right hand side is referred to as the *staggered charge*. We restrict our Hilbert states to the set of vectors $|\psi\rangle$ which satisfy equation $(\heartsuit)$, which also restricts to a subsector of fixed Q^V charge. Recalling that $[Q^V, Q^A] \neq 0$, we see that Q^A *cannot* be a symmetry of this theory, in a lattice analogue of the chiral anomaly. The Gauss' law for the axial gauge theory is completely analogous. We end on the note that, just as in the continuum theory how the Gauss law acts as a generator of $U(1)$ symmetries, we can define the Gauss' law operator [EZ18] of our theory as:

$$\mathcal{G}_j := E_{j+\frac{1}{2}} - E_{j-\frac{1}{2}} - q_j^V - \frac{(-1)^j}{2},$$

such that our physical states live in its kernel and, just as in the continuum, it generates the symmetries of the vector gauged Hamiltonian $\hat{H}_V$.

³¹We use q_j^V in our Gauss' law, whereas the Chatterjee-Pace-Shao paper uses $q_{j+\frac{1}{2}}^V$. It is the opinion of this author that this is a typo on their part.

5.2 We cannot gauge $U(1)^V$ and $U(1)^A$: The Onsager algebra

Recall from subsection 3.1 that the continuum charges Q^L and Q^R commute and hence $[Q^V, Q^A] = 0$. It is for this reason that the total global symmetry of the combined action is given by a direct group product $[U(1)^V \times U(1)^A] / \mathbb{Z}_2$. If this were true on the lattice, i.e. if $[Q^V, Q^A]$ vanished, we would have a similar combined action, and we could hope for another ultra-local charge description which could be gauged. Of course, we have already seen that this commutator is non-vanishing, and so we have no such path to gauging; the story of this obstruction to gauging is that of the Onsager algebra. Our non-vanishing commutator establishes that the combined algebra cannot simply be decomposed into $\mathfrak{u}(1)^V \oplus \mathfrak{u}(1)^A$, and so we prompt the question: What does our algebra look like? One of the novelties of [CPS25] is their identification of the Onsager algebra as the answer to our question. Introduced by the eponymous Lars Onsager, this Lie algebra was introduced in [Ons44] as part of an exact solution to the 2D Ising model. They define the following pair of operator families:

$$G_n := \frac{i}{2} \sum_{j=1}^L (a_j a_{j+n} - b_j b_{j+n}) \quad \text{and} \quad Q_n := \frac{i}{2} \sum_{j=1}^L a_j b_{j+n}, \quad \text{where } n \in \mathbb{Z}.$$

We can identify these general forms with operators we've already seen, for example:

$$\bullet Q^V = \frac{i}{2} \sum_{j=1}^L a_j b_j = Q_0, \quad \bullet Q^A = \frac{i}{2} \sum_{j=1}^L a_j b_{j+1} = Q_1, \quad \bullet [Q^V, Q^A] = iG_1.$$

Proposition 5.1. *These families of operators form the closed algebra:*

$$[Q_n, Q_m] = -iG_{m-n}, \quad [G_n, G_m] = 0, \quad [Q_n, G_m] = 2i(Q_{n-m} - Q_{n+m}).$$

Moreover, these operator families all commute with $\hat{H}$.

Proof. To begin with, we show that each of these operator families commute with our Hamiltonian $\hat{H}$. We note that the operator G_n splits into real (a) and imaginary (b) parts $G_n = G^a \oplus G^b$, and such we only need deal with $\hat{H}_a$ and $\hat{H}_b$ individually. Moreover, each derivation is the same up to labelling (and a sign), so we only need check $[\hat{H}_a, G_n^a]$:

$$\begin{aligned} [\hat{H}_a, G_n^a] &= \frac{1}{4} \sum_{j,l=1}^L [a_j a_{j+1}, a_l a_{l+n}] = \frac{1}{4} \sum_{j,l=1}^L 2 \left(-\delta_{j-l-n}^\nu a_l a_{j+1} + \delta_{l-j}^\nu a_{l+n} a_{j+1} - \delta_{j+1-l-n}^\nu a_j a_l + \delta_{l-j-1}^\nu a_j a_{l+n} \right) \\ &= \frac{1}{2} \sum_{j=1}^L (-a_j a_{j+n+1} + a_{j+n} a_{j+1} - a_{j+n} a_{j+1} + a_j a_{j+1+n}) = 0. \end{aligned}$$

So all the G s commute nicely, what about the Q s? We cannot use the same separation trick as such, but we will not write out the details for the b part.

$$[\hat{H}_a, Q_n] = \frac{1}{4} \sum_{j,l=1}^L [a_j a_{j+1}, a_l b_{l+n}] = \frac{1}{2} \sum_{j,l=1}^L (\delta_{l-j-1}^\nu a_j - \delta_{l-j}^\nu a_{j+1}) b_{l+n} = \frac{1}{2} \sum_{j=1}^L (a_j b_{j+n+1} - a_{j+1} b_{j+n}).$$

The form for $\hat{H}_b$ is the same, and the total commutator is given by:

$$[\hat{H}, Q_n] = \frac{1}{2} \sum_{j=1}^L (a_j b_{j+n+1} - a_{j+1} b_{j+n} - a_j b_{j+n+1} + a_{j+1} b_{j+n}) = 0.$$

Now, it is just a matter of verifying the form of the closed algebra:

$$\begin{aligned}
[Q_n, Q_m] &= -\frac{1}{4} \sum_{j,l=1}^L [a_j b_{j+n}, a_l b_{l+m}] = -\frac{1}{2} \sum_{j,l=1}^L (\delta_{l-j}^\nu b_{l+m} b_{j+n} - \delta_{l+m-j-n}^\nu a_j a_l) \\
&= \frac{1}{2} \sum_{j=1}^L (a_j a_{j+(n-m)} - b_j b_{j+(n-m)}) = -i G_{n-m}. \\
[G_n^a, G_m^a] &= -\frac{1}{4} \sum_{j,l=1}^L [a_j a_{j+n}, a_l a_{l+m}] \\
&= -\frac{1}{2} \sum_{j,l=1}^L (-\delta_{j-l-m}^\nu a_l a_{j+n} + \delta_{j-l}^\nu a_{l+m} a_{j+n} - \delta_{j+n-l-m}^\nu a_j a_l + \delta_{l-j-n}^\nu a_j a_{l+m}) \\
&= -\frac{1}{2} \sum_{j=1}^L (-a_j a_{j+n+m} + a_j a_{j+n-m} - a_j a_{j+n-m} + a_j a_{j+n+m}) = 0. \\
[Q_n, G_m] &= -\frac{1}{4} \sum_{j,l=1}^L [a_j b_{j+n}, a_l a_{l+m} - b_l b_{l+m}] \\
&= -\frac{1}{2} \sum_{j,l=1}^L ((\delta_{j-l}^\nu a_{l+m} - \delta_{j-l-m}^\nu a_l) b_{j+n} - a_j (\delta_{j+n-l}^\nu b_{l+n} - \delta_{j+n-l-m}^\nu b_l)) \\
&= -\sum_{j=1}^L (a_j b_{j+(n-m)} - a_j b_{j+(n+m)}) = 2i(Q_{n-m} - Q_{n+m}).
\end{aligned}$$

□

The central point is not only that this is a closed algebra, but rather that this Onsager algebra is *the* algebraic closure of Q^V and Q^A , and it features operators Q_n for arbitrarily large n . As such, we certainly do *not* find ultra-local operators, and so any attempt to gauge the combined charges will fail.

5.3 The mixed and the chiral anomalies vis-à-vis Nielsen-Ninomiya

The stories of the mixed $U(1)^V, U(1)^A$ anomaly and the chiral anomaly in the Schwinger model are intricately linked. In fact, the chiral anomaly emerges from combining the mixed anomaly with the requirement that $\partial_\mu j_V^\mu = 0$. In this context, the chiral anomaly appears as:

$$\partial_\mu j_A^\mu = \left(\sum_{\text{right}} q_A q_V - \sum_{\text{left}} q_A q_V \right) \frac{E}{2\pi}.$$

The derivation of this can be found in [Kap22]. We can see precisely here how the mixed $q_A q_V$ anomaly mediates the chiral anomaly. We have seen how, on the lattice, a mixed obstruction has emerged. Moreover, this mixed obstruction has emerged in a way with no analogue of anomaly cancellation appearing; we cannot opt to charge our fields differently. This inevitability of the mixed obstruction is the result of the conspiracy of the Nielsen-Ninomiya theorem. For each field with vector charge q_V , we are guaranteed that the spectrum on that charge sector has an equal number of left and right movers, and so:

$$\sum_{\text{right}} q_A q_V - \sum_{\text{left}} q_A q_V = q_V \sum_{\text{right}} (+1) - q_V \sum_{\text{left}} (-1) = 2q_V \times (\text{number of left-movers}).$$

The takeaway is this: Nielsen-Ninomiya enforces the mixed anomaly for any theory where the fields have the similarly-signed vector charge. In particular this is true for our theory where q_V is always -1 . This should be interpreted as bolstering the notion that, rather than it being a shame that Q^V and Q^A do not commute on the lattice, their non-local Onsager algebra actually enforces the mixed obstruction that is required by Nielsen-Ninomiya.

Remark 5.2. Our final note here is not as central as the above but still of interest: while Nielsen-Ninomiya bolsters the mixed anomaly, it is also the mechanism behind the individual anomaly cancellations. By this we mean that, within a charge sector q_V , we must have:

$$\sum_{\text{right}} q_V^2 = \sum_{\text{left}} q_V^2,$$

and overall each q_V^2 in the left sum has a copy in the right sum which cancels it out, and the same sort of story holds for q_A .

6 Conclusion

6.1 Essay reprise

As we draw the curtains on our story, we take a moment to reflect on the discussion we've been going through regarding the form and functionality of the axial charge operator introduced by Chatterjee, Pace, and Shao. We took a lattice Hamiltonian defined in $1+1$ -dimensions, defined as a tight-binding model with an imaginary hopping term, and we explicitly constructed the continuum low-energy effective theory in the limit where the lattice becomes infinite. This theory is explicitly one of right- and left-moving fermion fields, and from its momentum and charge conjugation operators, we have bridged the ideas of axial charge in the continuum and in the lattice theories.

The trials and tribulations rendered unto us by Nielsen and Ninomiya have been discussed in detail. We saw how, in the axial charge construction of the paper, the Nielsen-Ninomiya theorem takes on a life beyond being a no-go for chiral fermions, but rather acts as a guide connecting anomalies on the lattice and in the continuum.

We've put mathematical legs on the analysis present in the original paper and demonstrated explicitly how the lattice axial charge flows to the continuum axial charge in the infinite-extent limit. What's more, we've described the process of gauging this symmetry and that of the vector charge operator, with no appearance of self-anomalies, but seeing how gauging one renounces the other as a symmetry. Through our discussion of the Onsager algebra, we identify how the non-commutativity of our axial and vector symmetries provides an obstruction to gauging, which lends us an analogue to anomaly matching. The 't Hooft anomaly matching condition is an exact identification between the 't Hooft anomalies of a theory in the UV and in the IR. In our lattice theory, we observe an analogous effect: we identify an obstruction in the UV lattice theory with a 't Hooft mixed anomaly in the IR continuum theory.

6.2 Some final thoughts, and alternative avenues

We should be mindful of the parts of our discussion in which we have either been cavalier with our physics, or where there is more discussion to be had. It is not obvious from our construction how best to adapt these techniques to handle other gauge theories, be they in higher dimensions or different in their field composition. In particular the pairing between the mixed and chiral anomalies might be specific to our dimensionality, and it's not clear how it would scale up.

It's worth mentioning that our continuum effective theory isn't a one-to-one match with the theory we wanted to find, in particular because the operators χ_R and χ_L we introduced are not Majorana fermions as

they should be. This was a deliberate decision in order to most easily construct the Hamiltonian we were after. One way to remedy this for a more comprehensive construction would be the following definition:

$$\chi_R(x) := \frac{1}{\sqrt{V}} \sum_{0 \leq q < \frac{L}{2\pi} \Gamma} \left(e^{2\pi i x(q/L)} \beta_q + e^{-2\pi i x(q/L)} \beta_{-q} \right),$$

but this does not play well with our Fermi sea picture. It is possible that there is a better choice we did not think of. We would also like to have a more rigorous conversation about the precise nature of the $L \rightarrow \infty$, $\Gamma \rightarrow 0$ limit. Specifically, we have not made it clear at what rate these limits should be simultaneously taken.

There is much more to be said about Nielsen-Ninomiya, but one of the most relevant is in the myriad other ways it can be dealt with. Consider, for example, the “3450” chiral gauge theory, where we have two left-movers with vector charges of 3 and 4 and two right-movers with vector charges of 5 and 0. This theory has a cancelled vector anomaly in the continuum, but still cannot be put on the lattice naïvely because of Nielsen-Ninomiya. However, if we allow for non-quadratic interaction terms in our Hamiltonian, as is done in [WW19], we can find a lattice implementation of this chiral gauge theory. Moreover, an attempt has already been made to use the Chatterjee-Pace-Shao result to find a lattice theory whose continuum EFT is the “3450” theory in [Xu25], and this does not make use of these higher-order interaction terms. It should also be noted that this paper has spawned a resurgence of interest in the Onsager algebra [AKT26][GT26][Xu25]. It is not immediately obvious at the moment what, if any, deeper physical significance this algebra has. Certainly, before this result, it was considered a niche algebra for some exactly solvable models, so it is definitely of note that it has cropped up in a new context.

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Appendices

Appendix A: What lattice Hamiltonians are made of: a brief overview

We devote some attention here to a brief overview of hopping terms in lattice Hamiltonians. When writing down a Lagrangian for a classical continuum theory, we only include terms which respect the desired symmetries of our theory. Typical terms are very well studied and form building blocks of a theory which can be easily identified with physical aspects of our theory such as kinetic energy, mass energy, potentials, interactions. This symmetry respecting building block strategy holds up in much the same way when writing down lattice Hamiltonian theories. Because we are after a theory of free fermions, we need our theory to conserve particle number and to be invariant under lattice translations, which implies that $\hat{H}$ commutes with the particle number operator $\hat{n}$, the translation operator $\hat{T}_\beta$ which translates the field by the Bravais vector $\beta \in \mathbb{Z}^d$. Within the second-quantisation formalism, we can express these in terms of the field operators: $\hat{n} = \sum_\alpha c_\alpha^\dagger c_\alpha$, $\hat{T}_\beta = \sum_\alpha c_{\alpha+\beta}^\dagger c_\alpha$.

Each of the terms in $\hat{H}$ will also be borne of the field operators. The simplest term which commutes with $\hat{n}$ is the on-site potential term $\sum_\alpha V(\alpha) c_\alpha^\dagger c_\alpha$, which adds a potential energy $V(\alpha)$ to each site α , but this only commutes with $\hat{T}_\beta$ for constant V , which is a boring potential. In fact, the simplest non-trivial term which commutes with our number operator, translation operator, and the proper rotation operators is given by $\sum_{i=1}^d (\hat{T}_{\hat{\mu}_i} + \hat{T}_{-\hat{\mu}_i}) = \sum_{i=1}^d \hat{T}_{\hat{\mu}_i} + \text{h.c.}$, which is the result of extending $\hat{T}_{\hat{\mu}_1}$ to be isotropic. In single-particle QM, we can view such a Hamiltonian as a simple model of a wavefunction spreading over time, where at each time step an equal probability current flows in each direction.

Kinetic terms are essential for any semblance of dynamics to appear in our theory, and are also the most immediately questionable terms to include when working on the discrete lattice as they are typically taken to be derivatives of the field. Just as in condensed matter theories, we will liberate particles from single lattice sites via so-called *hopping terms*, which take the form of our earlier Hamiltonian term $t \sum_{i=1}^d (\hat{T}_{\hat{\mu}_i} + \text{h.c.})$, where we’ve also included a coupling t called the *hopping parameter*.

Remark. The connection between hopping terms and continuum kinetic energy is elucidating (and interesting in its own right). For simplicity, let’s say we’re dealing with a massive theory. Consider the symmetric discrete derivative

$$\Delta_i f(x) := \frac{1}{2a} (f(x + a\hat{\mu}_i) - f(x - a\hat{\mu}_i)) = \frac{1}{2a} (\hat{T}_{\hat{\mu}_i} - \hat{T}_{-\hat{\mu}_i}) f(x).$$

We define a discrete momentum analogue as $\pi_i = i\hbar\Delta_i$, and consider what action it generates:

$$\exp\left(-\frac{i}{\hbar}\pi_i a\right) = \exp(a\Delta_i) = \exp\left(\frac{1}{2}(\hat{T}_{\hat{\mu}_i} - \hat{T}_{-\hat{\mu}_i})\right).$$

From here, we use the formal generating function of the Bessel functions:

$$\exp\left(\frac{z}{2}(t - t^{-1})\right) = \sum_{n=-\infty}^{+\infty} J_n(z) t^n \implies \exp\left(-\frac{i}{\hbar}\pi_i a\right) f(x) = \sum_{n=-\infty}^{+\infty} J_n(1) \hat{T}_{\hat{\mu}_i}^n f(x).$$

The Bessel term $J_0(1)$ dominates all others, and we can say that

$$\exp\left(-\frac{i}{\hbar}\pi_i a\right) f(x) \approx \hat{T}_{\hat{\mu}_i} f(x),$$

and so we see the analogy: $\hat{p}$ generates continuum translations, π_i generates lattice translations. From this point, we can connect the kinetic terms:

$$t\left(\hat{T}_{\hat{\mu}_i} + \hat{T}_{-\mu_i}\right) \approx t\left(\exp\left(-\frac{i}{\hbar}\pi_i a\right) + \exp\left(+\frac{i}{\hbar}\pi_i a\right)\right) = t\left(2 - \frac{a^2}{\hbar^2}\pi_i^2 + \mathcal{O}(a^4)\right).$$

Choosing $t := -\frac{\hbar^2}{2ma^2}$ and taking the limit as $a \rightarrow 0$, we find:

$$t\left(\hat{T}_{\hat{\mu}_i} + \hat{T}_{-\mu_i}\right) \rightarrow \frac{\pi_i^2}{2m},$$

just as is in the continuum theory.

Appendix B: Rotation operators on the lattice

This section is more general than our work requires, dealing with an arbitrarily high-dimensional lattice. This would be useful in the extension of this work to a discussion of 1 + 3-dimensional QED. Chirality and rotation are intrinsically linked. Of particular importance for our work are the improper rotations, which will interchange left-moving and right-moving operators in the continuum limit. We therefore devote some time to a detailed explanation of how rotation works in these systems, whose complications demand extra attention anyway.

Thinking of the lattice classically, the group which describes the discrete symmetries of the space is given by the hyperoctohedral group $\mathbb{Z}_2^d \wr S_d$. This wreath product group acts on the unit vectors $\hat{\mu}^i$ by permuting the actions of the d simple reflections $\alpha_1 \rightarrow -\alpha_1, \alpha_1 \rightleftharpoons \alpha_2, \dots, \alpha_{d-1} \rightleftharpoons \alpha_d$, where α_i is the i -th co-ordinate. Notice that each simple reflection is an improper rotation, but the combination of any two is a proper rotation. We could describe just the proper rotations with $\mathbb{Z}_2^d \wr A_d$ where $A_d \trianglelefteq S_d$ is the subgroup of even permutations. We would like to manifest this action in the Hamiltonian formalism such that the rotation operator $\hat{R}_r$ takes an element $r \in \mathbb{Z}_2^d \wr S_d$ which takes an operator $c_\alpha^\dagger$ and returns $\sim c_{r(\alpha)}^\dagger$. Here we will explicitly show how to implement the simple reflection operators and we claim that any composite rotation formed by permutation can be built using permutation matrices. This will suffice for our 1 + 1-dimensional lattice where we only have the simple reflection $\alpha_1 \rightarrow -\alpha_1$.

Simple reflection I $\alpha_1 \rightarrow -\alpha_1$:

This rotation is analogous to a parity symmetry P in just the first dimension. We will call this a *flip* and define the following flip charge operator:

$$\hat{F} := \sum_{j=1}^{L/2} \hat{f}_j, \text{ where } \hat{f}_j := i(c_j^\dagger c_{L-j} - c_{L-j}^\dagger c_j).$$

We've suppressed all of the other spatial indices for clarity. Equivalently we can see that we're restricting our analysis to the first grade of our direct sum space, that is to say just the lattice. We claim that each of the component operators $\hat{f}_j$ can be considered as charge operators for a $U(1)$ action on our space. Combined with the observation that different $\hat{f}_j$'s commute, this justifies our naming of $\hat{F}$ as a charge operator. We will use this fact and prove this fact in that order.

Note the following:

$$\begin{aligned} [\hat{f}_j, c_i] &= i \left(c_j^\dagger c_{L-j} c_i - c_{L-j}^\dagger c_j c_i - c_i c_j^\dagger c_{L-j} + c_i c_{L-j}^\dagger c_j \right) \\ &= i(\delta_{i+j-L}^\nu c_j - \delta_{i-j}^\nu c_{L-j}), \end{aligned}$$

from which it readily follows that $[\hat{f}_j, c_j] = -ic_{-j}$ and $[\hat{f}_j, c_{L-j}] = ic_j$. We see that $\text{ad}_{\hat{f}_j}$ acts on $\langle c_j, c_{L-j} \rangle_{\mathbb{C}}$ as σ^y , whose diagonalisation we know to be $c_j - ic_{L-j}$ and $c_j + ic_{L-j}$ with eigenvalues $+1$ and -1 respectively. It follows immediately that $\text{Ad}_{e^{i\theta\hat{F}}}(c_j \mp ic_{L-j}) = e^{i\theta\text{ad}_{\hat{F}}}(c_j \mp ic_{L-j}) = e^{\pm i\theta}(c_j \mp ic_{L-j})$. Moving back into our basis, we see that this rotation acts as $\text{Ad}_{e^{i\theta\hat{F}}}(c_j) = \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right) c_j + \left(\frac{e^{i\theta} - e^{-i\theta}}{2i} \right) c_{L-j} = \cos(\theta)c_j + \sin(\theta)c_{L-j}$. So our quantisation of this discrete simple reflection has resulted in a $U(1)$ interpolation between superposed states. Picking $\theta = \pi/2$, we get exactly the result we'd hoped for: $c_j \mapsto c_{L-j}$. All this follows mutatis mutandis for $c_j^\dagger$ and $c_{L-j}^\dagger$, but to show that this operator is suitable for our whole operator algebra $\mathcal{A}_j \cong \text{End}(|\text{vac}\rangle, |j\rangle, |L-j\rangle, |j, L-j\rangle)_{\mathbb{C}}$, we follow through with our earlier claim that $\hat{f}_j$ is a charge operator for a $U(1)$ action. This requires showing that $\text{ad}_{\hat{f}_j}$ has quantised eigenvalues.

Proof. We do this by expounding upon our earlier observation that $\text{ad}_{f_j} \sim \sigma^y$, and consider operators the $\hat{e}_j := c_j^\dagger c_{L-j} + c_{L-j}^\dagger c_j$ and $\hat{g}_j := c_j^\dagger c_j - c_{L-j}^\dagger c_{L-j}$ such that $e_j, f_j, g_j \sim \sigma^x, \sigma^y, \sigma^z$. We have hence identified an $\mathfrak{su}(2)$ subalgebra of $\mathcal{A}_j$ under which the embedding $\text{ad} : \mathfrak{su}(2) \hookrightarrow \text{Aut}(\mathcal{A})$ is automatically a Lie algebra representation. Because $\mathcal{A}$ is finite-dimensional, we can decompose $\mathcal{A}$ into irreps in the standard way whence we find that the weights given by eigenvalues of $\text{ad}_{\hat{f}_j}$ are quantised. $\square$

Simple reflection II $\alpha_k \rightleftharpoons \alpha_{k+1}$:

This rotation is only slightly trickier to implement than the first simple reflection. By analogy, we can correctly guess that our operator should be:

$$\hat{S} := \sum_{i=1, j \geq i}^{i=L} \hat{s}_{i,j}, \text{ where } \hat{s}_{i,j} := i \left(c_{i,j}^\dagger c_{j,i} - c_{j,i}^\dagger c_{i,j} \right).$$

Once again we suppress all indices that are not in the k -th or $k+1$ -th positions. We refer to this operator as the *swap* charge operator. All of our arguments for the flip charge operator carry over to this operator, so we won't repeat them here.

Remark. Having written down these rotation operators, a subtlety of our model comes into the limelight as it becomes very clear that we have not included the idea of spin into our model, despite the fact that we are modelling fermions. This is acceptable in our discussion of [CPS25] because this takes place in a staggered-fermionic lattice, which has already spread the spin-degrees of freedom spatially. Because our operators are written in terms of fermion bilinears, we could repeat all of the above arguments for bosonic field operators and come to the same conclusions. Within the Hamiltonian lattice formalism that we have laid out, with no discussion of Lorentz symmetry, we have no spin-statistics theorem. This is consistent with other spinless/integer spin fermion models in condensed matter theory. For example, in the Heisenberg spin model, we have bosonic spin operators which can be effectively *fermionised* through the Jordan-Wigner transformation.

Appendix C: Derivation of quotient of $[U(1)^\nu \times U(1)^{\mathcal{A}}] / \mathbb{Z}_2$

The combined action of $U(1)^\nu \times U(1)^{\mathcal{A}}$ is not faithful however, as $e^{i\pi Q^\nu} = e^{i\pi Q^L} e^{i\pi Q^R} = e^{i\pi Q^L} e^{-i\pi Q^R} = e^{i\pi Q^A}$. To get the faithful action, we need to quotient out by the kernel of the group homomorphism

$\rho : U(1) \rightarrow \text{Aut}(\mathcal{A})$ where $\mathcal{A}$ is the operator algebra generated by $\{\Psi_L, \Psi_R, \Psi_L^\dagger, \Psi_R^\dagger\}$:

$$\ker \rho = \left\{ \left(e^{i\phi_1 \mathcal{Q}^V}, e^{i\phi_2 \mathcal{Q}^A} \right) : e^{i\phi_1 q_V} e^{i\phi_2 q_A} = 1 \right\}.$$

It follows readily that $\phi_1 q_V + \phi_2 q_A = 0 \pmod{2\pi}$ in the above expression, and we know that $q_V, q_A \in \{-1, +1\}$, from which it follows that $\phi_1 = \pi n_V, \phi_2 = \pi n_A$ for $n_V, n_A \in \mathbb{Z}$. Our equation then becomes $q_A n_A + q_V n_V = 0 \pmod{2}$, which simplifies to $n_A + n_V = 0 \pmod{2}$ when we recall that $-1 = +1 \pmod{2}$. This equation has solution set $(n_1, n_2) \in \{(0, 0), (1, 1)\} \pmod{2}$, hence $\ker \rho$ is the $\mathbb{Z}_2$ subgroup given by $\{(1, 1), (e^{i\pi \mathcal{Q}^V}, e^{i\pi \mathcal{Q}^A})\}$, meaning our faithful action comes from the group $[U(1)^V \times U(1)^A] / \mathbb{Z}_2$.

Appendix D: Semi-direct product structure

Proposition. *The larger global symmetry group can be expressed as the semi-direct product $U(1)^\mathcal{L} \rtimes_\varphi \mathbb{Z}_2^{\mathcal{L}}$, where φ is the group homomorphism from $\mathbb{Z}_2^{\mathcal{L}}$ to $\text{Aut}(U(1)^\mathcal{L})$ where $\varphi_d(\phi) = (-1)^d \phi$ for $d \in \mathbb{Z}_2^{\mathcal{L}}$.*

Proof. Consider some $(e^{i\phi}, d) \in U(1) \times \mathbb{Z}_2$.³² If we let G act on Ψ^L through conjugation, first via $(\mathcal{C}^L)^d$, and then via $e^{i\phi \mathcal{Q}^L}$, we find:

$$\begin{aligned} e^{i\phi \mathcal{Q}^L} (\mathcal{C}^L)^d \Psi_L (\mathcal{C}^L)^{-d} e^{-i\phi \mathcal{Q}^L} &= e^{i\phi \mathcal{Q}^L} \begin{Bmatrix} \Psi_L & d=0 \\ \Psi_L^\dagger & d=1 \end{Bmatrix} e^{-i\phi \mathcal{Q}^L} = \begin{Bmatrix} e^{-i\phi} \Psi_L & d=0 \\ e^{+i\phi} \Psi_L^\dagger & d=1 \end{Bmatrix} \\ &= (\mathcal{C}^L)^d \begin{Bmatrix} e^{-i\phi} \Psi_L & d=0 \\ e^{+i\phi} \Psi_L & d=1 \end{Bmatrix} (\mathcal{C}^L)^{-d} \\ &= (\mathcal{C}^L)^d e^{(-1)^d i\phi \mathcal{Q}^L} \Psi_L e^{-(-1)^d i\phi \mathcal{Q}^L} (\mathcal{C}^L)^{-d}. \end{aligned}$$

Mutatis mutandis for $\Psi_L^\dagger$ yields:

$$e^{i\phi \mathcal{Q}^L} (\mathcal{C}^L)^d \Psi_L^\dagger (\mathcal{C}^L)^{-d} e^{-i\phi \mathcal{Q}^L} = (\mathcal{C}^L)^d e^{(-1)^d i\phi \mathcal{Q}^L} \Psi_L^\dagger e^{-(-1)^d i\phi \mathcal{Q}^L} (\mathcal{C}^L)^{-d}.$$

These formulae describe how to commute the two symmetries past one another, meaning we can always write any combination of actions of these group elements as one action of $\mathbb{Z}_2^{\mathcal{L}}$ followed by one action of $U(1)^\mathcal{L}$, which we denote: $(\exp i\phi, d)(\Psi_L)$ or $(\exp i\phi, d)(\Psi_L^\dagger)$. We've already shown that³³ $(\exp i\phi, d) = (\exp i\phi, 0) \circ (1, d) = (1, d) \circ (\exp [(-1)^d i\phi], 0)$, from which we can deduce that:

$$\begin{aligned} (\exp i\phi_1, d_1) \circ (\exp i\phi_2, d_2) &= (\exp i\phi_1, 0) \circ (1, d_1) \circ (\exp i\phi_2, 0) \circ (1, d_2) \\ &= (\exp i\phi_1, 0) \circ (\exp [(-1)^{d_1} i\phi_2], 0) \circ (1, d_1) \circ (1, d_2) = (\exp [\phi_1 + (-1)^{d_1} \phi_2], d_1 + d_2), \end{aligned}$$

which is precisely the group multiplication law for the semi-direct product $U(1)^\mathcal{L} \rtimes_\varphi \mathbb{Z}_2^{\mathcal{L}}$ for the group homomorphism φ defined above. $\square$

The semi-direct structure given via φ is the unique non-Abelian semi-direct product $U(1)^\mathcal{L} \rtimes \mathbb{Z}_2^{\mathcal{L}}$, and so we can identify our group as a presentation $\cong O(2)$. All of the above reasoning applies just as well for the right-moving action S_R , in which we similarly define a $U(1)^{\mathcal{R}}$ symmetry with a $\mathcal{Q}^R$ charge operator.

³²This is as a Cartesian product, and should not be interpreted as a group direct product.

³³Strictly speaking we have not shown this, as this is assuming our action is faithful. We can work with this w.l.o.g., however, as we can always descend an action to a faithful action by quotienting out its kernel.